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CHANGES IN THE U.S. WAGE STRUCTURE 1963–1987: APPLICATION OF QUANTILE REGRESSION

BY MOSHE BUCHINSKY¹

The 1980's witnessed rapid changes in the structure of wages in the United States. In this study, recently developed techniques for quantile regression are applied to every March Current Population Survey since 1964 and changes in the return to schooling and experience at different points of the wage distribution are examined. The quantile regression technique parsimoniously describes the entire conditional wage distribution. This conditional distribution is used to examine changes in within-group wage inequality as measured by the difference between conditional quantiles. Two linear models are considered: a simple one-group model, and a 16-group model. The results suggest that the returns to schooling and experience differ across quantiles of the wage distribution but their patterns of change are similar. Significant differences in wage inequality are also found across the various skill groups. The paper also presents a new imputation method for CPS data.

KEYWORDS: Quantile regression, iterative linear programming algorithm (ILPA), bootstrap estimator, order statistic estimator, Current Population Survey (CPS).

1. INTRODUCTION

THE 1980'S WITNESSED RAPID AND MASSIVE CHANGES in the structure of wages in the United States. In particular one observes sharp changes in wage inequality, and dramatic increases in wage differentials by education and by experience. Various aspects of these changes in the wage structure have attracted much recent research and are well documented.² Most studies in this area share several common observations. Wage inequality declined during the 1960's, most noticeably for blacks in comparison to whites but also within racial groups. The inequality appears to have increased since the mid-1970's, and the trend accelerated during the 1980's. The returns to education and labor market experience changed enormously during this period and played a major role in widening the wage inequality.

Competing explanations for these phenomena differ greatly. One common explanation is that the demand for labor has shifted from "highly-skilled" to "less skilled" workers. Technical changes have been posited as one reason for

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² See, for example, studies by Blackburn, Bloom, and Freeman (1990, 1991), Bound and Johnson (1991, 1992), Juhn, Murphy, and Pierce (1993), Katz and Murphy (1992), Katz and Revenga (1989), Murphy and Welch (1989, 1992). For an excellent survey of studies in this area see Levy and Murnane (1992).

these demand shifts (Davis and Haltiwanger (1991), Bound and Johnson (1992), Krueger (1991), Mincer (1991)). Another explanation stresses the trade deficit, which led to the decline in the manufacturing sector and in turn to greater demand for more “highly-educated” workers (Murphy and Welch (1991)). Changes in wage-setting institutions, such as the decline in unionization and the fall in the real value of minimum wage, have been stressed by others as important factors (Blackburn, Bloom, and Freeman (1990, 1991), Freeman (1991), and Mitchell (1989)).

The changes in wage structure have also been attributed to: (a) the growth of “more-educated” workers; (b) amount of schooling among different racial groups; and (c) the quality of schools (Blackburn, Bloom, and Freeman (1991), Card and Krueger (1992a, 1992b)). Katz and Murphy (1992) conclude that labor supply fluctuations combined with stable demand growth, explain much, but certainly not all, of the changes in education and wage differentials.

In this paper I do not offer new explanations, nor do I try to refute any of the existing hypotheses. Rather, I apply recently developed techniques for quantile regression analysis to parsimoniously describe the entire conditional wage distribution.³ I examine changes in the returns to skills—in particular, education and experience—at different points of the wage distribution. I also construct and analyze two measures of inequality which, because of the quantile regression method used, depend on the model’s covariates.

Several practical problems are dealt with in this study. I develop a number of techniques that facilitate the empirical application of quantile regression and apply them to every March Current Population Survey (March CPS) since 1964. In particular, new estimators for the asymptotic covariance matrix are used, and a new algorithm for estimating the censored quantile regression model is introduced. I also propose and utilize a new imputation method for variables which are reported only on a bracketed basis before 1976.

Two models are considered: (a) a simple one-group model; and (b) a 16-group model, fitted for narrowly defined education-experience groups. The results for the one-group model show that in general the returns to education and experience are quite different at the various quantiles. Nevertheless, the changes over time have a similar pattern for all the quantiles. However, when a finer 16-group model is examined the returns at different quantiles do not exhibit the same pattern of changes for all skill groups. Moreover, the pattern of changes at any quantile is very different for the various skill groups.

Wage inequality—defined as the difference between two conditional quantiles—changed in a different manner for the skill groups. Furthermore, the

³ Chamberlain (1991) uses a similar technique in examining other aspects of the changes in the U.S. wage structure between 1979 and 1987. The study by Juhn, Murphy, and Pierce (1993) is also very close in spirit to quantile regression analysis as they look at changes in various quantiles of the wage distribution for narrowly defined skill groups. In an earlier paper, Wholstetter and Coleman (1972) also examine income disparities between whites and nonwhites (in the U.S.) by comparing quantiles from their wage distributions, controlling for differences in age, occupation, and schooling for the two groups.

changes toward the end of the sample period come mainly from longer tails at the low end of the wage distribution, especially for the less educated workers.

The paper is organized as follows. Section 2 describes the data. Section 3 presents the econometric models and estimation methods employed, and also points out potential problems. Section 4 discusses the empirical results of the two models. The 16-group model serves as a benchmark for comparing the results of the parsimonious one-group model. Section 5 offers a summary and conclusions. An appendix presents the imputation method for the March CPS data.

2. THE DATA

The analysis in this paper is based on data for over a million people drawn from the March Current Population Survey (March CPS) for 1964 through 1988. The March CPS sampling frame is the civilian population of the U.S. living in housing units.

The extract used here—consisting of between 10,000 and 34,000 observations for each year (see Table VI)—contains all black and white males between the ages of 18 and 70, which typically accounts for about 96% of the sample of males. The sample is also restricted to individuals: (a) who worked at least one week in the year prior to the March CPS year; (b) whose current jobs were either in the private or in the government sector (excluded are persons who were self-employed, working without pay, or who had never worked); and (c) who earned at least \$50 per week in 1982 prices.⁴ Occasional instances where further selection criteria are imposed are explicitly noted.

Some of the questions in the March CPS refer to activity in the previous year. Hence the dates in this paper, 1963 through 1987, correspond to March CPS years 1964 through 1988. Nominal data are deflated by the implicit price deflator of personal consumption expenditures for gross national product.⁵

Prior to 1975 some of the March CPS variables are reported only on a bracketed basis; for example, the variable “weeks worked last year” is reported only in seven categories. Several methods have been used in the literature to impute values for such variables. The most common method assigns to each individual the mean value of a group with the same characteristics from the post 1975 period. Although this method, when applied to CPS data, yields imputed variables that are highly correlated with the true variables, in many cases the point estimates derived from mean and quantile regressions that use the true and imputed variables, differ by more than two standard errors. One reason for this difference is that reported earnings are usually rounded, and therefore have “spikes” at several levels. If we assign the same value of weeks worked to a

⁴ This selection criterion is comparable to those used in most of the recent literature on this subject. One should be cautious, however, in interpreting the results because of possible selection bias introduced by this procedure. This is undoubtedly a topic for future research.

⁵ Taken from The Economic Report of the President, 1990.

large fraction of the population, we preserve these “spikes” and thereby induce smaller variation in the dependent variable.

Since the main variable of interest in this study is the weekly wage, defined by the “total income from wages and salaries last year” divided by “[the number of] weeks worked last year,” it is important to accurately impute the latter variable. In this study I propose an imputation method which seems to be more accurate than the usual method. The procedure uses the conditional distributions of weeks worked (conditional on race, age, and education) in the years 1975, 1976, and 1977 to impute values for each individual in the years prior to 1975. In general, this method provides imputed values which are highly correlated (around .98–.99) with the true values for 1978 and later. Point estimates derived from regressions (similar to those for the one-group and 16-group models) which use the imputed values, differ by less than one standard error from those using the true values. This is important since the imputation has been conditioned on only three covariates—there are more covariates in the actual regressions. A more detailed description of this imputation method is provided in the Appendix.

An important, previously recognized, problem is that some of the data for earnings are top coded for confidentiality. A common practice among labor economists (e.g., Murphy and Welch (1992), Katz and Revenga (1989), and Juhn, Murphy, and Pierce (1993) is to multiply the top coded value by 1.45, implicitly assuming that the earnings distribution above the censoring point of \$999 has a Pareto tail, so that the log of earnings has an exponential distribution whose mean is proportional to the censoring point. The value 1.45 is determined from calculations performed on years in which there is practically no censoring.⁶

I address this problem by resorting to econometric models which use information only on observations that are not censored. This method, which is discussed further in the next section, also helps to overcome problems associated with changes in sampling method, coverage, and imputation techniques, that occurred in the 1976 March CPS.

3. METHODOLOGY

This section discusses the econometric models used in this study and provides some intuitive interpretations. It also presents alternative estimators for the asymptotic covariance matrix of the estimated coefficients.

The quantile regression model of Koenker and Bassett (1978) can be written as

$$y_i = x_i' \beta_\theta + u_{\theta i} \quad \text{with} \quad \text{Quant}_\theta(y_i | x_i) = x_i' \beta_\theta \quad (i = 1, \dots, n),$$

where β_θ and x_i are $K \times 1$ vectors, and $x_{i1} \equiv 1$. $\text{Quant}_\theta(y|x)$ denotes the θ th

⁶ In fact, it is likely that this value has increased over time.

conditional quantile of y given x . Also, let $f_{u_\theta}(\cdot|x)$ denote the density of u_θ given x .⁷

3.1. Quantile Regression—Estimation by Minimum Distance

If the distribution of the covariate vector x has finite support, with $\Pr(x = x^j) = \phi_j$ ($j = 1, \dots, J$), we can obtain a minimum distance (MD) estimator for β_θ (e.g., Chamberlain (1991)). The MD estimator, which solves $\min_\beta (\hat{Q}_\theta - G\beta)'A^{-1}(\hat{Q}_\theta - G\beta)$, is given by $\hat{\beta}_\theta = (G'A^{-1}G)^{-1}G'A^{-1}\hat{Q}_\theta$, where $Q'_\theta = (Q_\theta^1, \dots, Q_\theta^J)$, $Q_\theta^j \equiv \text{Quant}_\theta(y|x = x^j)$, $\hat{Q}_\theta$ is the estimate of Q_θ , G is a $J \times K$ matrix ($J \geq K$) with rows x^j ($j = 1, \dots, J$), and $A \xrightarrow{p} \Psi$, a positive-definite matrix. A standard MD result is that $\sqrt{N}(\hat{\beta}_\theta^{MD} - \beta_\theta) \xrightarrow{d} \mathcal{N}(0, \Lambda_\theta^{MD})$, where

$$(1) \quad \Lambda_\theta^{MD} = (G'\Psi^{-1}G)^{-1}G'\Psi^{-1}\Lambda_{Q_\theta}\Psi^{-1}G(G'\Psi^{-1}G)^{-1},$$

$\Lambda_{Q_\theta} = \text{diag}(\sigma_{\theta 1}^2/\phi_1, \dots, \sigma_{\theta J}^2/\phi_J)$ is the asymptotic covariance of $\hat{Q}_\theta$, and $\sigma_{\theta j}^2 = \theta(1-\theta)/f_{u_\theta}^2(0|x = x^j)$. An efficient estimator for β_θ is obtained by taking $A = \hat{\Lambda}_{Q_\theta}$, where $\hat{\Lambda}_{Q_\theta}$ is a consistent estimator of Λ_{Q_θ} .⁸

One can regard the linear specification of $\text{Quant}_\theta(y|x)$ as merely an approximation for a more complex function. We can still define the parameter vector of the linear predictor by $\beta_\theta \equiv \arg \min_\beta E[\rho_\theta(y - x'\beta)]$, where $\rho_\theta(\lambda) = (\theta - I(\lambda < 0))\lambda$, and $I(\cdot)$ is the usual indicator function. Letting $Q_\theta^* \equiv G\beta_\theta$ denote the population linear predictor of the conditional quantiles Q_θ , it follows that

$$(2) \quad \begin{aligned} \sqrt{N}(\hat{\beta}_\theta - \beta_\theta) &= (G'A^{-1}G)^{-1}G'A^{-1}\sqrt{n}(\hat{Q}_\theta - Q_\theta) \\ &\quad + \sqrt{N}\left((G'A^{-1}G)^{-1}G'A^{-1} - (G'\Psi^{-1}G)^{-1}G'\Psi^{-1}\right)(Q_\theta - G\beta_\theta). \end{aligned}$$

Chamberlain (1991) showed that for $A^{-1} = \text{diag}(N_1/N, \dots, N_J/N)$ (N_j is the number of observations with $x = x^j$), $\sqrt{N}(\hat{\beta}_\theta - \beta_\theta) \xrightarrow{d} \mathcal{N}(0, \Lambda_\theta^{MD} + \Lambda_\theta^M)$, where Λ_θ^{MD} , given in (1), corresponds to the first term in (2), and

$$\begin{aligned} \Lambda_\theta^M &= (G'\Psi^{-1}G)^{-1}G'\Psi^{-1} \text{diag}(\delta_{\theta 1}^2/\phi_1, \dots, \delta_{\theta J}^2/\phi_J) \\ &\quad \times \Psi^{-1}G(G'\Psi^{-1}G)^{-1}, \end{aligned}$$

with $\delta_{\theta j} = Q_{\theta j} - \beta_\theta'x^j$, corresponds to the second term in (2).⁹

⁷ It follows that $\text{Quant}(u_\theta|x) = 0$. It is assumed that the distribution function of u_θ given x , $F_{u_\theta}(\cdot|x)$, is continuously differentiable with density function $f_{u_\theta}(0|x) > 0$.

⁸ In Buchinsky (1991) I showed that this estimator attains the semiparametric efficiency bound of Newey and Powell (1990) for quantile regression. In their case the estimator solves $\min_\beta \sum_{i=1}^N f_{u_\theta}(0|x_i)\rho_\theta(y_i - x_i'\beta)$, and x need not be discrete.

⁹ Since $Q_\theta \neq G\beta_\theta$, the rate of convergence of $\sqrt{N}(\hat{\beta}_\theta - \beta_\theta)$ depends on the rate of convergence of the elements in A . It will converge at a rate slower than $\sqrt{N}$ if one uses the "optimal" matrix $A = \hat{\Lambda}_{Q_\theta}$ for efficiency gain. Buchinsky (1994a) provides the limiting distribution of a similar linear predictor estimator for the more general case of the Box-Cox transformation model.

Censoring problems can easily be dealt with in an intuitive way in this framework. In particular, in estimating β_θ we use only those cells for which the estimates of the conditional quantiles $\hat{Q}_\theta^j$ are below the censoring value.

3.2. Quantile Regression—The General Case

Koenker and Bassett (1978) suggested the quantile regression estimator for β_θ (where x need not be discrete) to be a solution to

$$(3) \quad \min_{\beta} \frac{1}{N} \sum_{i=1}^N \rho_\theta(y_i - x_i' \beta).$$

The population θ th quantile of y , μ_θ , is a solution to $\min_{\mu} E[\rho_\theta(y - \mu)]$.¹⁰ The *analogy principle* (e.g., Manski (1988)) suggests the sample analog, $\hat{m}_\theta$, as a solution to $\min_m (1/N) \sum_{i=1}^N \rho_\theta(y_i - m)$. Since $\text{Quant}_\theta(y|x) = x' \beta_\theta$, a solution to $\min_{\mu} E[\rho_\theta(y - \mu)|x]$ is $\mu_\theta = x' \beta_\theta$. Thus β_θ is a solution to $\min_{\beta} E[\rho_\theta(y - x' \beta)]$, and by the analogy principle $\hat{\beta}_\theta$ is a solution to (3).

Powell (1984, 1986) showed that the problem in (3) fits into the generalized method of moments (GMM) framework for the censored quantile regression model, under Huber's (1967) regularity conditions, and therefore we have $\sqrt{N}(\hat{\beta}_\theta - \beta_\theta) \rightharpoonup \mathcal{N}(0, \Lambda_\theta)$,¹¹ where

$$(4) \quad \Lambda_\theta = \theta(1 - \theta) \left(E[f_{u_\theta}(0|x)xx'] \right)^{-1} E(xx') \left(E[f_{u_\theta}(0|x)xx'] \right)^{-1}.$$

If we assume that the density of u_θ is independent of x , i.e., $f_{u_\theta}(0|x) = f_{u_\theta}(0)$, as is assumed by Koenker and Bassett (1978), then Λ_θ simplifies to

$$(5) \quad \Lambda_\theta = \frac{\theta(1 - \theta)}{f_{u_\theta}^2(0)} (Exx')^{-1} = \sigma_\theta^2 (Exx')^{-1},$$

where $\sigma_\theta^2 = \theta(1 - \theta)/f_{u_\theta}^2(0)$.

Linear Programming Representation

Problem (3) can be shown to be a linear programming (LP) problem. In this study the Barrodale and Roberts (1973) algorithm, which employs the simplex method, is used to solve (3).

Two features of the LP representation are important here: (a) problem (3) is the *primal problem* of LP; and (b) the first order conditions (F.O.C) satisfy the

¹⁰ The population θ th quantile is defined by $\mu_\theta \equiv \inf\{\mu: F_y(\mu) \geq \theta\}$, where F_y is the c.d.f. of y . If y is continuous, then $\theta = \Pr(y \leq \mu_\theta) = F_y(\mu_\theta)$, and $\mu_\theta \equiv F_y^{-1}(\theta)$. The sample quantile is defined similarly by $\hat{m}_\theta \equiv \inf\{m: F_{N,y}(m) \geq \theta\}$, where $F_{N,y}(\cdot)$ is the empirical distribution of $y_1, \dots, y_N$.

¹¹ The F.O.C. for problem (3) are $(1/N) \sum_{i=1}^N (\theta - 1/2 + 1/2 \text{sgn}(y_i - x_i' \beta)) x_i = o_p(N^{-1/2})$, where $\text{sgn}(\lambda) = I(\lambda \geq 0) - I(\lambda \leq 0)$. That is, the F.O.C. cannot hold exactly.

conditions for the *dual problem* of LP.¹² The Duality Theorem of LP implies then that an optimal solution exists if $X = (x_1, \dots, x_N)'$ has a full column rank. Moreover, the solution is obtained in a finite number of simplex iterations.

An important implication of the LP problem is that y_i can be increased toward ∞ if $y_i - x_i'\hat{\beta}_\theta > 0$, or y_i can be decreased toward $-\infty$ if $y_i - x_i'\hat{\beta}_\theta < 0$, without altering the solution $\hat{\beta}_\theta$. This is of signal importance for the current study, since the March CPS data are censored.¹³

3.3. Censored Quantile Regression

The censored quantile regression model was proposed by Powell (1984, 1986). The results in some of the analyses here are appreciably affected by the censoring.

In the latent variable setup we can write the model as follows. Let $y_i^* = x_i'\beta_\theta + u_{\theta i}$ with $\text{Quant}_\theta(u_{\theta i}|x_i) = 0$, and let $y_i = y_i^*$ if $y_i^* \leq y^0$ and $y_i = y^0$ if $y_i^* > y^0$, where y^0 is the censoring value. Then the conditional quantile of y , given x , is given by $\text{Quant}_\theta(y|x) = \min(y^0, x'\beta_\theta)$.

A consistent estimator for β_θ is obtained as a solution to

$$(6) \quad \min_{\beta} \frac{1}{N} \sum_{i=1}^N \rho_\theta(y_i - \min(y^0, x_i'\beta))$$

where $\rho_\theta(\lambda)$ is as in (3). The F.O.C. for (6) are given by

$$(7) \quad \frac{1}{N} \sum_{i=1}^N I(x_i'\hat{\beta}_\theta < y^0) \left(\theta - 1/2 + 1/2 \text{sgn}(y_i - x_i'\hat{\beta}_\theta) \right) x_i = o_p(N^{-1/2}).$$

That is, in estimating β_θ only the observations for which $x_i'\hat{\beta}_\theta < y^0$ are used.¹⁴ Powell (1984, 1986) showed that $\sqrt{N}(\hat{\beta}_\theta - \beta_\theta) \xrightarrow{d} \mathcal{N}(0, \Lambda_\theta^P)$, where

$$(8) \quad \Lambda_\theta^P = \theta(1 - \theta) \left(E \left[f_{u_\theta}(0|x) I(x'\beta_\theta < y^0) x x' \right] \right)^{-1} E \left[I(x'\beta_\theta < y^0) x x' \right] \\ \times \left(E \left[f_{u_\theta}(0|x) I(x'\beta_\theta < y^0) x x' \right] \right)^{-1}.$$

¹² The Equilibrium Theorem of LP implies that there exists an $N \times 1$ vector w with: (i) $w_i = \theta$ if $\hat{u}_{\theta i} > 0$; (ii) $w_i = -(1 - \theta)$ if $\hat{u}_{\theta i} < 0$; and (iii) $-(1 - \theta) \leq w_i \leq \theta$ if $\hat{u}_{\theta i} = 0$, such that $\sum_{i=1}^N w_i x_i = 0$, where $\hat{u}_{\theta i} \equiv y_i - x_i'\hat{\beta}_\theta$. The terms $w_i = (\theta - 1/2 + 1/2 \text{sgn}(\hat{u}_{\theta i}))$ from the F.O.C. coincide with those implied by the LP solution in cases (i) and (ii). For (iii), however, the F.O.C. give $(\theta - 1/2 + 1/2 \text{sgn}(\hat{u}_{\theta i})) = 0$, while the dual problem specifies $-(1 - \theta) \leq w_i \leq \theta$. This shows the precise sense in which the F.O.C. hold approximately (see footnote 11). See also Bartels and Conn (1980).

¹³ Juhn, Murphy, and Pierce (1993) showed that changes in the CPS sampling introduced in 1976 mainly affected the upper and lower 10% of the sample. To the extent that this coincides with the upper and lower 10% of the distribution of u_θ , conditional on the characteristics x , $\hat{\beta}_\theta$ should not be affected by these changes as long as $.1 \leq \theta \leq .9$.

¹⁴ Dropping the observations for which $x_i'\hat{\beta}_\theta \geq y^0$ is of no consequence since $y_i - \min(y^0, x_i'\hat{\beta}_\theta) = y_i - y^0$, which does not depend on $\hat{\beta}_\theta$. Also, similar to the quantile regression case, the F.O.C. in (7) holds only approximately (see footnotes (11) and (12)).

Similar to the quantile regression model, if $f_{u_\theta}(0|x) = f_{u_\theta}(0)$ then Λ_θ^P simplifies to

$$(9) \quad \Lambda_\theta^P = \frac{\theta(1-\theta)}{f_{u_\theta}^2(0)} \left(E \left[I(x'\beta_\theta < y^0) xx' \right] \right)^{-1} = \sigma_\theta^2 \left(E \left[I(x'\beta_\theta < y^0) xx' \right] \right)^{-1}.$$

Estimation of the Censored Quantile Regression

Since $\min(y^0, x'_i\beta)$ is not linear in β , (6) is not an LP problem. Nonetheless, an LP algorithm can be used in an iterative manner as follows. In the j th iteration solve for $\hat{\beta}_\theta^{(j)}$ using the observations for which $x'_i\hat{\beta}_\theta^{(j-1)} < y^0$. Stop when the sets of observations in two consecutive iterations are the same. In this algorithm—which I call the *iterative linear programming algorithm* (ILPA)—the number of iterations for convergence to occur need not be finite. If, however, the number is finite, then a local minimum is guaranteed.¹⁵ Typically, for the March CPS data set, convergence has always occurred in 5 to 20 iterations regardless of the starting value $\hat{\beta}_\theta^0$ used. This is an indication that a *global* minimum is obtained.¹⁶

3.4. *Estimation of the Asymptotic Covariance*

Two alternative methods are employed in estimating the asymptotic covariance matrix for $\hat{\beta}_\theta$ in the quantile and censored quantile regression models: (a) the *order statistic* (OS) estimator, for covariance matrices given by (5) and (9); and (b) the *design matrix bootstrap* (DMB) estimator, for more general covariance matrices given by (4) and (8).

The OS estimator is used in the computation of the asymptotic covariance matrix $\hat{\Lambda}_{Q_\theta}$ for $\hat{Q}_\theta$. This matrix is then used in the computation of the adjusted asymptotic covariance matrix of $\hat{\beta}_\theta$ for the one-group model, as outlined in Section 3.1.

The DMB estimator is used in the computation of the asymptotic covariance matrix for the coefficient estimates for the 16-group model. The DMB method used here provides valid estimates for the asymptotic covariance matrix of the quantile estimates even if one considers the quantile regression estimates only as linear predictors.¹⁷

¹⁵ For a proof see Buchinsky (1991). Womersley (1986) suggested an alternative algorithm which is an extension of the Barrodale-Roberts algorithm for the censored case. Womersley presents a compact characterization of the generalized gradient for $s_N(\beta) = \sum_{i=1}^N |y_i - \max(y^0, x'_i\beta)|$, and gives necessary and sufficient conditions for strict local minimization of $s_N(\beta)$. The advantage of Womersley's algorithm is that it guarantees convergence to a solution in a finite number of iterations (which is not guaranteed by the ILPA algorithm) since the method reduces the search to a finite number of points β . In practice the ILPA algorithm provides no problems and convergence to a solution was always achieved.

¹⁶ In all the estimations in this study I use the median regression estimate, with the constant properly adjusted, as the starting value for β_θ .

¹⁷ See, for example, Hahn (1992), Buchinsky (1992), and the Monte Carlo experiments reported in Buchinsky (1994b).

Order Statistic Estimator—The Independence Case¹⁸

For this case we can estimate σ_θ^2 in (5) by matching two confidence intervals for the population quantile μ_θ —an exact confidence interval from a binomial distribution and a confidence interval using a normal approximation. More specifically, from the binomial distribution and its limit for large N

$$(10) \quad \Pr(U_{(s)} \leq \mu_\theta \leq U_{(t)}) = \Pr(s \leq X_B < t) \xrightarrow{P} \Phi\left(\frac{+l}{\sqrt{N\theta(1-\theta)}}\right) - \Phi\left(\frac{-l}{\sqrt{N\theta(1-\theta)}}\right),$$

where $X_B \sim B(N, \theta)$, $U_{(m)}$ denotes the sample m th order statistic, $s = [N\theta - l]$, $t = [N\theta + l]$, and $[\lambda]$ is the integer part of λ . Denoting by Z_q the q th quantile of a standard normal variable, the normal approximation implies

$$(11) \quad \Pr(\hat{\mu}_\theta - \sigma_\theta N^{-1/2} Z_{1-\alpha/2} \leq \mu_\theta \leq \hat{\mu}_\theta + \sigma_\theta N^{-1/2} Z_{1-\alpha/2}) \xrightarrow{P} \Phi(Z_{1-\alpha/2}) - \Phi(-Z_{1-\alpha/2}).$$

Equating the two confidence intervals from (10) and (11) gives

$$\hat{\sigma}_\theta^2 = N(\hat{U}_{(t)} - \hat{U}_{(s)})^2 / 4Z_{1-\alpha}^2, \quad \text{and} \quad l = Z_{1-\alpha/2} \sqrt{N\theta(1-\theta)},$$

where $\hat{U}_{(m)}$ is the m th order statistic of $\hat{u}_{\theta 1}, \dots, \hat{u}_{\theta N}$. An estimate $\hat{\Lambda}_\theta$ is then provided by $\hat{\Lambda}_\theta = \hat{\sigma}_\theta^2 ((1/N) \sum_{i=1}^N x_i x_i')^{-1}$.¹⁹

The OS estimator is in essence a histogram estimator of $f_{u_\theta}(0)$. The derivative of the error term's quantile function is $dF_{u_\theta}^{-1}(\theta)/d\theta = 1/f_{u_\theta}(F_{u_\theta}^{-1}(\theta))$. The order statistic estimator uses differences between two order statistics to estimate this derivative. The corresponding estimator for $f_{u_\theta}(0)$ can be written in histogram form as $(F_{N\hat{u}_\theta}(\hat{u}_{(t)}) - F_{N\hat{u}_\theta}(\hat{u}_{(s)}))/(\hat{u}_{(t)} - \hat{u}_{(s)})$, where $\hat{u}_{(s)}$ is the s th order statistic and $F_{N\hat{u}_\theta}$ is the empirical distribution of $\hat{u}_{\theta i}$ ($i = 1, \dots, N$).

Design Matrix Bootstrap Estimator—The General Case

For the general case, where $f_{u_\theta}(0|x) \neq f_{u_\theta}(0)$, the bootstrap estimator is employed, using the general method suggested by Efron (1979, 1982).²⁰ B

¹⁸ This estimator was suggested by Huber (1981) and was also used by Chamberlain (1991).

¹⁹ For the censored quantile regression model an estimate $\hat{\sigma}_{\theta P}^2$ for σ_θ^2 is based only on the observations for which $x_i' \hat{\beta}_\theta < y_i^0$, and $\hat{\Lambda}_\theta^P = \hat{\sigma}_{\theta P}^2 ((1/N) \sum_{i=1}^N I(x_i' \hat{\beta}_\theta < y_i^0) x_i x_i')^{-1}$.

²⁰ In general, there is little theoretical justification for the use of the bootstrap method in econometric models. The bootstrap method for quantile regression was first implemented by Buchinsky (1991) and was adopted also by Chamberlain (1991). Several authors considered bootstrapping the sample quantile (e.g., Efron (1979, 1982), Bickel and Freedman (1981), Singh (1981), and Lo (1989)). Hahn (1991, 1992) proved the consistency of the bootstrap estimator for quantile regression and for a class of M -estimators, using the percentile method (see Efron (1982)) rather than the method employed here. Buchinsky (1992) justified the use of the bootstrap method for quantile regression for the method employed here, but only for discrete x . Freedman (1981) provided justification for the same bootstrapping scheme used here for mean regression. Freedman and Peters (1984a, 1984b) looked at some empirical results of bootstrapping econometric models, but they provided no theoretical justification.

samples are drawn, each of size m (m need not be equal to N), from the (x_i, y_i) pairs. For each of the B samples an estimator for β_θ is computed to obtain B bootstrap estimates, $\hat{\beta}_\theta^1, \dots, \hat{\beta}_\theta^B$. The estimate of Λ_θ is computed then by²¹

$$(12) \quad \hat{\Lambda}_\theta = N \left(\frac{m}{N} \right) \frac{1}{B} \sum_{b=1}^B (\hat{\beta}_\theta^b - \hat{\beta}_\theta) (\hat{\beta}_\theta^b - \hat{\beta}_\theta)'$$

Instead of using $\hat{\beta}_\theta$ as the pivotal vector in (12) one can use the average of the bootstrap estimates, say $\hat{\beta}_\theta^A$. There are two reasons for using $\hat{\beta}_\theta$ as the pivotal vector: (a) the bootstrap method is based on the idea that the conditional law of $\sqrt{m}(\hat{\beta}_\theta^b - \hat{\beta}_\theta)$ (conditional on the sample $(x_i, y_i), i = 1, \dots, N$) is "close" to the law of $\sqrt{N}(\hat{\beta}_\theta - \beta_\theta)$; and (more importantly) (b) the method used here has better small sample properties as is shown in Buchinsky (1994b).

This last study reports on a large scale Monte Carlo investigation of various estimation methods. The "population" for this study is the 1987 CPS sample of all outgoing rotation groups of white males (75,578 observations). "Population parameter" vectors at five quantiles (.10, .25, .50, .75, and .90) were computed for the same linear model employed here. Monte Carlo samples of varying sizes were drawn from the CPS "population." For each sample and each estimation method the model parameters and .95 nominal confidence intervals were estimated. For the bootstrap estimator several different bootstrap sample sizes were used. Each experiment was repeated 1000 times, and the fraction of times that the nominal confidence intervals contained the population parameter value (i.e., the empirical level) was computed for each parameter.

The results of this study exhibit excellent performance of the bootstrap method; the empirical levels for the bootstrap estimators are within less than a half standard error from the .95 nominal level. Furthermore, the bootstrap method is robust at all quantiles to changes in the bootstrap sample size (m) relative to the data sample size (N).

4. EMPIRICAL RESULTS

Two models are considered in this empirical section: a simple one-group model, and a 16-group model. The former contains only few discrete explanatory variables and therefore can be estimated by the MD framework. The second model has more variables and hence a linear programming algorithm (more precisely the ILPA algorithm) must be employed. Several considerations bear on the application of these models: (a) in some empirical applications only the simple one-group model can be employed; (b) the one group model places more restrictions on the functional form than the 16-group model; comparison of the two models indicates what features are needed for a flexible one-group model; and (c) the one-group model is much simpler computationally; one

²¹ The resampling schemes for the quantile and censored quantile regression models are essentially the same. The only difference is the manner in which the bootstrap estimates are obtained. The term m/N in (12) accounts for the fact that $m \neq N$.

should weigh the computation complexity of the 16-group model against the advantage of having more detailed information.

The quantile regression model allows one to estimate the entire conditional distribution of y given x . While more quantile regressions can potentially be more informative, estimation is restricted here to five quantiles: .10, .25, .50, .75, and .90.²² The dependent variable in all regressions is the logarithm of weekly wages, defined by total annual income from wages and salaries divided by number of weeks worked.

The analysis employs a reduced form equation and emphasizes mainly the returns to education and experience, and especially the patterns of change in these returns. The term “return to experience” (and likewise “return to education”) refers merely to the effect of experience on the conditional quantile of log wage. The causal effect interpretation is beyond the scope of this study. Consequently, I do not address identification questions related to wage equations of the type used here (e.g., Heckman and Robb (1985)).²³

4.1. *Simple Quantile Regression Model—One-Group Model*

In this section a linear model of the type introduced by Mincer (1974), is estimated for the entire sample. In this model the explanatory variables are: a constant, education, experience, experience squared, and a dummy variable for blacks.²⁴ Each quantile regression is estimated for each of the sample years, 1963 to 1987, using the MD framework.²⁵ The estimation uses only cells with: (a) 15 or more observations; (b) at least 7 years of education (the reason will become clearer after the results of the 16-group model are presented); and (c) the conditional quantile below the censoring value.²⁶

4.1.1. *One-Group Model—Return to Education*

The estimated returns to education at the five quantiles are reported, in percentages, in Table I. Table I also reports the mean return to education from

²² It might be of concern to estimate a .10 quantile if in fact a large fraction of the data on earnings were at the minimum wage level. This potential problem is irrelevant here since all the .10 conditional quantiles estimated are always above the minimum wage level.

²³ This is not to say that the identification problem raised by Heckman and Robb is irrelevant. However, it is the usual practice in the literature to call the coefficient on education “returns to education” (and similarly for experience).

²⁴ Education is defined by: Education = highest grade attended $- 1 - I$ (if last grade attended was not completed). Experience—actually, potential experience—is defined by: Experience = $\min(\text{age} - \text{education} - 6, \text{age} - 18)$. Both education and experience are measured in single years. The dummy variable for blacks takes the value 1 if an individual is black and 0 otherwise.

²⁵ As Murphy and Welch (1990) point out, this functional form is quite restrictive for such a wide range of ages. However, the MD framework limits us to a simple functional form. It is also of empirical interest to check whether this functional form is uniformly restrictive across all quantiles.

²⁶ Only the estimation of the .75 and .90 quantile regressions are affected by censoring. The years in which censoring matters are: 1969, 1973 through 1987 at the .90 quantile, and 1979, 1980, 1982, and 1983 at the .75 quantile (see Table VI).

TABLE I
ONE-GROUP MODEL—RETURN TO EDUCATION
MEAN, .10, .25, .50, .75, .90 QUANTILES, AND RESTRICTED ESTIMATES
DEPENDENT VARIABLE: LOG OF TOTAL WEEKLY INCOME FROM WAGES AND SALARIES

Year	Regressions						Restricted
	Mean	.10 Qnt.	.25 Qnt.	.50 Qnt.	.75 Qnt.	.90 Qnt.	
1963	6.65 (0.22)	6.45 (0.54)	6.23 (0.45)	6.35 (0.29)	6.58 (0.37)	6.84 (0.42)	6.48 (0.25)
1964	6.51 (0.22)	5.07 (0.57)	6.44 (0.48)	6.41 (0.30)	6.46 (0.34)	7.22 (0.43)	6.39 (0.25)
1965	6.84 (0.13)	6.81 (0.42)	6.82 (0.27)	6.60 (0.16)	6.61 (0.20)	7.48 (0.31)	6.67 (0.15)
1966	6.85 (0.19)	6.90 (0.55)	6.02 (0.42)	6.44 (0.24)	7.04 (0.29)	7.88 (0.48)	6.73 (0.22)
1967	6.82 (0.13)	5.91 (0.40)	6.54 (0.27)	6.57 (0.17)	6.88 (0.19)	7.74 (0.33)	6.72 (0.15)
1968	6.53 (0.13)	5.98 (0.42)	6.19 (0.25)	6.26 (0.16)	6.82 (0.19)	7.30 (0.31)	6.45 (0.15)
1969	7.05 (0.14)	5.99 (0.43)	6.48 (0.27)	6.89 (0.17)	7.52 (0.21)	8.28 (0.28)	7.14 (0.16)
1970	7.32 (0.14)	6.21 (0.44)	6.44 (0.30)	7.18 (0.18)	7.81 (0.21)	8.50 (0.31)	7.43 (0.16)
1971	7.24 (0.14)	5.88 (0.45)	6.82 (0.31)	7.20 (0.18)	7.73 (0.21)	8.37 (0.31)	7.38 (0.16)
1972	7.35 (0.15)	6.48 (0.50)	6.92 (0.31)	7.35 (0.19)	7.80 (0.22)	8.53 (0.31)	7.55 (0.17)
1973	7.18 (0.15)	6.87 (0.49)	6.80 (0.29)	7.10 (0.19)	7.28 (0.22)	7.92 (0.31)	7.20 (0.17)
1974	6.95 (0.16)	5.92 (0.47)	6.33 (0.32)	7.00 (0.19)	7.59 (0.23)	8.37 (0.30)	7.22 (0.17)
1975	7.70 (0.14)	7.24 (0.48)	7.58 (0.28)	7.58 (0.17)	7.85 (0.21)	8.52 (0.29)	7.77 (0.16)
1976	7.26 (0.12)	7.26 (0.39)	7.20 (0.22)	7.21 (0.16)	7.37 (0.18)	7.42 (0.26)	7.30 (0.14)
1977	7.14 (0.12)	7.03 (0.40)	7.14 (0.25)	6.96 (0.16)	7.44 (0.18)	7.71 (0.26)	7.16 (0.14)
1978	6.96 (0.13)	6.86 (0.40)	7.14 (0.25)	6.91 (0.17)	7.07 (0.20)	7.59 (0.24)	7.11 (0.15)
1979	6.78 (0.12)	6.74 (0.37)	6.95 (0.21)	6.80 (0.15)	7.10 (0.17)	7.46 (0.22)	7.00 (0.13)
1980	6.86 (0.12)	6.46 (0.38)	6.91 (0.22)	6.99 (0.15)	7.55 (0.17)	7.06 (0.21)	7.14 (0.13)
1981	7.41 (0.13)	7.11 (0.40)	7.52 (0.24)	7.29 (0.17)	7.81 (0.19)	8.35 (0.25)	7.61 (0.15)
1982	8.30 (0.14)	7.91 (0.44)	8.39 (0.25)	8.15 (0.18)	8.77 (0.20)	9.31 (0.26)	8.49 (0.16)
1983	8.62 (0.13)	8.25 (0.38)	8.70 (0.24)	8.52 (0.18)	9.07 (0.20)	9.22 (0.24)	8.78 (0.15)
1984	8.90 (0.14)	8.72 (0.43)	8.95 (0.26)	8.80 (0.19)	9.23 (0.21)	9.70 (0.27)	9.10 (0.16)
1985	9.58 (0.14)	9.38 (0.43)	9.46 (0.26)	9.51 (0.18)	9.78 (0.21)	10.86 (0.26)	9.81 (0.16)
1986	9.72 (0.15)	9.08 (0.42)	9.69 (0.26)	9.52 (0.19)	9.96 (0.22)	11.05 (0.28)	9.90 (0.16)
1987	9.63 (0.14)	8.73 (0.41)	9.40 (0.26)	9.62 (0.18)	9.92 (0.21)	11.05 (0.25)	9.94 (0.16)

Note: The regressions are computed using the MD framework. Each regression includes a constant, education, experience, experience squared, and a dummy variable for blacks. The reported numbers are the coefficients on education times 100. In parentheses are the adjusted standard errors which account for possible misspecification as described in Section 3.1. In addition to the selection criteria of Section 2, the sample includes only cells with 15 or more observations (10 in 1963 and 1964) and 7 or more years of education. The restricted estimates are obtained by MD constraining the slope coefficients of the five unrestricted vectors to be the same.

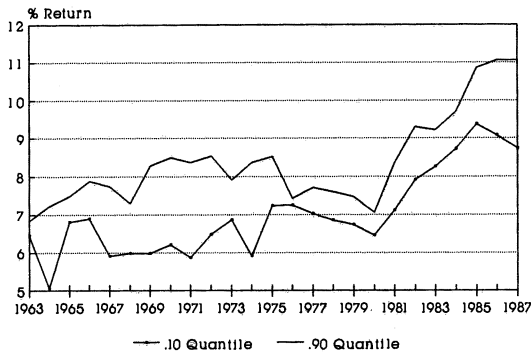
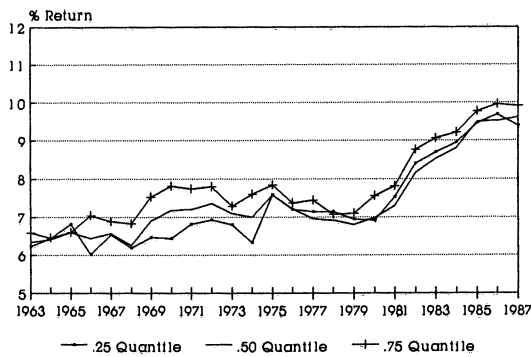
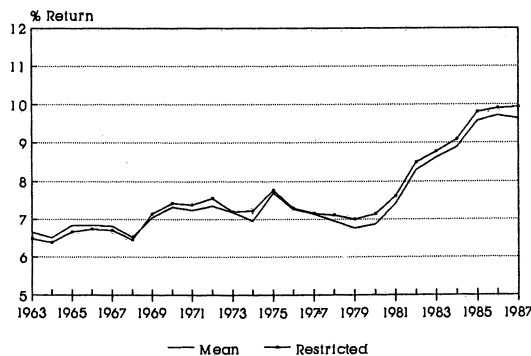
a. .10 and .90 Quantiles**b. .25, .50, and .75 Quantiles****c. Restricted across Quantiles and Mean**

FIGURE 1.—One-group model—return to education, by quantiles.

MD estimation using the sample means, and restricted estimates.²⁷ Test statistics for the null hypothesis of equality among all slope coefficients are reported in Table VI. The returns to education are also graphed in Figure 1.

The most striking result is that the mean return to education and the returns at each quantile changed in a similar pattern. However, the returns to education at these five quantiles differ significantly. In general the returns are higher at the higher quantiles (i.e., the upper end of the wage distribution), especially during the 1960's and the first half of the 1970's. During the second half of the 1970's when there was a general decline in the return to education, the returns at the five quantiles were similar.

During the 1980's, there were sharp unprecedented increases in the returns to education, with greater increases at the higher quantiles. For example, a mere 0.7 percentage difference between the returns at the two extreme quantiles (.90 and .10), increased to about 2.3 percentage by 1987. Furthermore, while the returns at the three highest quantiles increased until 1987, they decreased at the two lowest quantiles toward the end of the period. By 1987 the differences among the returns at the five quantiles were comparable to those in the early 1970's.

Dissimilarities can also be observed in the year-to-year fluctuations. The fluctuations at the middle quantiles (.25, .50, and .75) are smaller than at the extreme quantiles. This is due in part to sampling variation as is seen from the higher standard errors at the extreme quantiles than at the middle quantiles.

Finally, note that the mean estimates are very close to, though in general lower than, the restricted estimates. The two estimates demonstrate similar patterns of changes over time, and the differences between the two estimates are insignificant.

4.1.2. *One-Group Model—Return to Experience*

The return to experience—more precisely, the derivative of the conditional quantile of log weekly wage with respect to experience—is given by $\beta_2 + 2\beta_3 \times \text{Ex.}$, and needs to be evaluated at some specified level of experience.²⁸ Two points were chosen: 5 years of experience, representing new entrants; and 15 years of experience, representing experienced workers (I also refer to workers with 25 years of experience as prime earners). The results (in percentages) are reported in Table II, and graphed in Figure 2 for the new entrants and in Figure 3 for experienced workers.

A comparison of Figures 2 and 3 shows very different returns for the two experience groups. For the new entrants returns are in general higher at the

²⁷ The restricted coefficients are obtained by MD constraining the slope coefficients at the five quantiles to be the same. For a weight matrix the covariance matrix of the five unrestricted quantile estimates is used. Note that asymptotically, under the null hypothesis, $N(\beta - G_R \hat{\beta}_R) A^{-1} (\beta - G_R \hat{\beta}_R) \sim \chi^2(16)$, where $\hat{\beta}' = (\hat{\beta}'_{.10}, \dots, \hat{\beta}'_{.90})$, $\hat{\beta}'_R = (\hat{\beta}_{.10,1}, \dots, \hat{\beta}_{.90,1}, \hat{\beta}_2, \hat{\beta}_3, \hat{\beta}_4, \hat{\beta}_5)$, $A = \text{Va}(\hat{\beta})$, and G_R is a suitable restriction matrix.

²⁸ The coefficients β_2 and β_3 correspond to the coefficients on experience and experience squared, respectively. Ex. denotes experience.

TABLE II
ONE-GROUP MODEL—RETURN TO EXPERIENCE
MEAN, .10, .25, .50, .75, .90 QUANTILES, AND RESTRICTED ESTIMATES
DEPENDENT VARIABLE: LOG OF TOTAL WEEKLY INCOME FROM WAGES AND SALARIES

Regression	Mean	.10 Ont.	.25 Ont.	.50 Ont.	.75 Ont.	.90 Ont.	Restricted
Panel A: 5 Years of Experience							
Year							
1963	3.68 (0.13)	4.53 (0.36)	4.00 (0.26)	3.52 (0.17)	3.28 (0.20)	3.30 (0.24)	3.54 (0.14)
1964	3.88 (0.12)	4.39 (0.30)	4.09 (0.22)	3.86 (0.17)	3.47 (0.17)	3.74 (0.23)	3.79 (0.13)
1965	3.98 (0.08)	4.95 (0.23)	4.17 (0.15)	3.84 (0.10)	3.71 (0.12)	3.61 (0.17)	3.85 (0.08)
1966	3.91 (0.11)	4.46 (0.31)	3.94 (0.22)	3.62 (0.14)	3.67 (0.15)	3.56 (0.27)	3.70 (0.12)
1967	3.73 (0.08)	4.58 (0.22)	4.11 (0.15)	3.59 (0.11)	3.42 (0.11)	3.13 (0.19)	3.60 (0.09)
1968	3.57 (0.08)	4.43 (0.23)	3.71 (0.16)	3.40 (0.10)	3.38 (0.11)	3.25 (0.18)	3.45 (0.09)
1969	3.99 (0.09)	4.55 (0.24)	4.04 (0.15)	3.85 (0.10)	3.83 (0.12)	3.95 (0.18)	3.89 (0.09)
1970	4.25 (0.08)	4.56 (0.23)	4.30 (0.15)	4.05 (0.10)	4.06 (0.11)	4.24 (0.16)	4.12 (0.09)
1971	4.78 (0.08)	5.37 (0.22)	4.93 (0.16)	4.43 (0.10)	4.47 (0.11)	4.72 (0.17)	4.53 (0.09)
1972	4.82 (0.08)	5.42 (0.24)	4.99 (0.15)	4.63 (0.10)	4.67 (0.11)	4.80 (0.16)	4.69 (0.08)
1973	4.95 (0.08)	5.55 (0.26)	5.04 (0.15)	4.78 (0.09)	4.77 (0.11)	4.79 (0.17)	4.81 (0.08)
1974	4.85 (0.09)	5.28 (0.27)	4.79 (0.17)	4.73 (0.10)	4.59 (0.12)	4.86 (0.19)	4.71 (0.09)
1975	4.68 (0.08)	4.98 (0.22)	4.97 (0.14)	4.88 (0.09)	4.45 (0.11)	4.26 (0.17)	4.74 (0.08)
1976	4.82 (0.07)	5.31 (0.21)	5.05 (0.12)	4.93 (0.09)	4.63 (0.10)	4.33 (0.16)	4.83 (0.08)
1977	4.91 (0.07)	5.18 (0.19)	5.20 (0.13)	4.87 (0.09)	4.72 (0.10)	4.62 (0.14)	4.83 (0.08)
1978	4.69 (0.07)	4.64 (0.20)	4.85 (0.12)	4.94 (0.09)	4.65 (0.10)	4.48 (0.14)	4.79 (0.08)
1979	4.52 (0.07)	4.73 (0.18)	4.69 (0.11)	4.78 (0.08)	4.42 (0.09)	4.01 (0.13)	4.60 (0.07)
1980	4.54 (0.07)	4.56 (0.19)	4.70 (0.12)	4.80 (0.08)	4.56 (0.09)	4.24 (0.13)	4.65 (0.07)
1981	4.53 (0.08)	4.55 (0.20)	4.61 (0.14)	4.71 (0.09)	4.55 (0.11)	4.36 (0.15)	4.62 (0.08)
1982	4.63 (0.08)	4.43 (0.22)	4.68 (0.14)	4.95 (0.10)	4.80 (0.11)	4.44 (0.15)	4.82 (0.09)
1983	4.90 (0.08)	4.88 (0.22)	5.10 (0.15)	5.04 (0.10)	4.90 (0.11)	4.63 (0.15)	4.95 (0.09)
1984	5.02 (0.08)	5.02 (0.21)	5.30 (0.15)	5.09 (0.11)	4.93 (0.11)	4.93 (0.15)	5.03 (0.09)
1985	4.88 (0.08)	5.08 (0.23)	5.02 (0.14)	4.97 (0.10)	4.93 (0.11)	4.75 (0.17)	4.93 (0.09)
1986	4.94 (0.08)	5.06 (0.22)	5.23 (0.15)	5.05 (0.11)	4.94 (0.12)	4.75 (0.17)	4.99 (0.09)
1987	4.63 (0.08)	4.55 (0.21)	4.83 (0.15)	4.83 (0.11)	4.70 (0.12)	4.84 (0.16)	4.77 (0.09)

TABLE II—CONTINUED

Regression	Mean	.10 Qnt.	.25 Qnt.	.50 Qnt.	.75 Qnt.	.90 Qnt.	Restricted
Panel B: 15 Years of Experience							
Year							
1963	2.19 (0.07)	2.45 (0.19)	2.22 (0.12)	2.07 (0.09)	2.02 (0.10)	2.11 (0.12)	2.09 (0.07)
1964	2.28 (0.06)	2.34 (0.16)	2.33 (0.11)	2.26 (0.09)	2.12 (0.09)	2.31 (0.11)	2.23 (0.07)
1965	2.27 (0.04)	2.61 (0.11)	2.30 (0.08)	2.18 (0.05)	2.15 (0.06)	2.21 (0.09)	2.20 (0.04)
1966	2.21 (0.06)	2.40 (0.16)	2.14 (0.11)	2.02 (0.07)	2.10 (0.08)	2.21 (0.13)	2.08 (0.06)
1967	2.13 (0.04)	2.43 (0.11)	2.24 (0.08)	2.02 (0.06)	1.99 (0.06)	1.93 (0.09)	2.04 (0.05)
1968	2.03 (0.04)	2.33 (0.12)	2.02 (0.08)	1.90 (0.05)	1.94 (0.06)	1.96 (0.09)	1.93 (0.04)
1969	2.26 (0.04)	2.41 (0.12)	2.16 (0.08)	2.17 (0.05)	2.21 (0.06)	2.36 (0.09)	2.19 (0.04)
1970	2.39 (0.04)	2.41 (0.11)	2.30 (0.08)	2.28 (0.05)	2.34 (0.05)	2.52 (0.08)	2.33 (0.04)
1971	2.65 (0.04)	2.80 (0.11)	2.65 (0.08)	2.46 (0.05)	2.55 (0.05)	2.76 (0.08)	2.53 (0.04)
1972	2.73 (0.04)	2.91 (0.12)	2.76 (0.07)	2.61 (0.05)	2.68 (0.05)	2.81 (0.08)	2.65 (0.04)
1973	2.78 (0.04)	3.00 (0.12)	2.72 (0.07)	2.65 (0.05)	2.73 (0.05)	2.82 (0.08)	2.69 (0.04)
1974	2.73 (0.04)	2.85 (0.13)	2.60 (0.08)	2.65 (0.05)	2.64 (0.06)	2.86 (0.09)	2.66 (0.05)
1975	2.67 (0.04)	2.74 (0.11)	2.74 (0.07)	2.75 (0.05)	2.57 (0.05)	2.56 (0.08)	2.69 (0.04)
1976	2.75 (0.03)	2.93 (0.10)	2.81 (0.06)	2.80 (0.04)	2.68 (0.05)	2.55 (0.08)	2.76 (0.04)
1977	2.82 (0.03)	2.90 (0.09)	2.93 (0.06)	2.80 (0.04)	2.76 (0.05)	2.70 (0.07)	2.79 (0.04)
1978	2.71 (0.03)	2.61 (0.09)	2.77 (0.06)	2.82 (0.04)	2.68 (0.05)	2.62 (0.07)	2.75 (0.04)
1979	2.62 (0.03)	2.64 (0.09)	2.67 (0.05)	2.74 (0.04)	2.61 (0.05)	2.42 (0.06)	2.67 (0.03)
1980	2.64 (0.03)	2.52 (0.09)	2.69 (0.06)	2.80 (0.04)	2.70 (0.04)	2.52 (0.06)	2.72 (0.03)
1981	2.69 (0.04)	2.56 (0.10)	2.70 (0.06)	2.80 (0.04)	2.73 (0.05)	2.63 (0.07)	2.74 (0.04)
1982	2.78 (0.04)	2.57 (0.11)	2.79 (0.06)	2.97 (0.05)	2.88 (0.05)	2.73 (0.07)	2.90 (0.04)
1983	2.95 (0.04)	2.77 (0.11)	3.00 (0.07)	3.04 (0.05)	2.97 (0.05)	2.82 (0.07)	2.97 (0.04)
1984	3.01 (0.04)	2.90 (0.10)	3.11 (0.07)	3.06 (0.05)	3.00 (0.05)	3.00 (0.07)	3.03 (0.04)
1985	2.94 (0.04)	2.90 (0.11)	2.97 (0.07)	3.02 (0.05)	2.99 (0.05)	2.95 (0.08)	3.00 (0.04)
1986	3.00 (0.04)	2.88 (0.11)	3.11 (0.07)	3.08 (0.05)	3.02 (0.05)	2.97 (0.08)	3.05 (0.04)
1987	2.82 (0.04)	2.55 (0.10)	2.88 (0.07)	2.94 (0.05)	2.90 (0.05)	2.99 (0.07)	2.91 (0.04)

Note: See note in Table I.

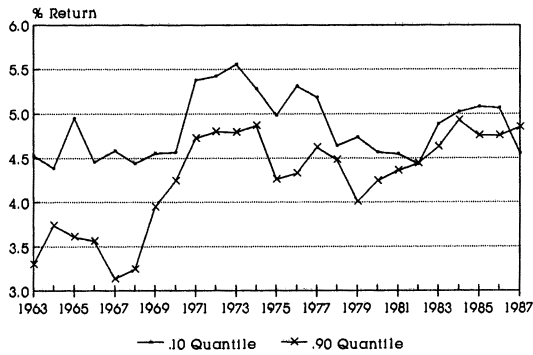
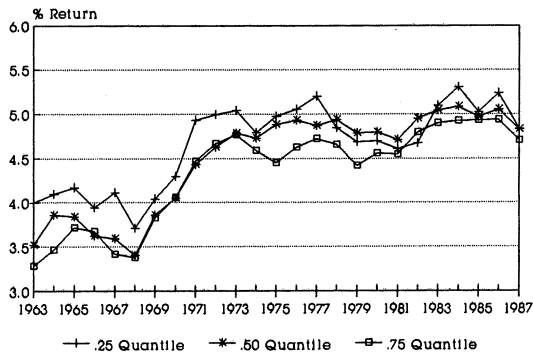
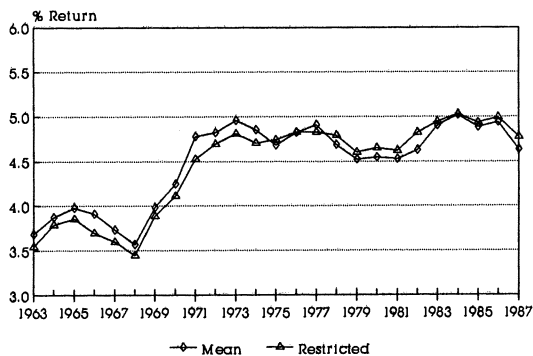
a. .10 and .90 Quantiles**b. .25, .50, and .75 Quantiles****c. Restricted across Quantiles and Mean**

FIGURE 2.—One-group model—return to experience for new entrants (5 years of experience), by quantiles.

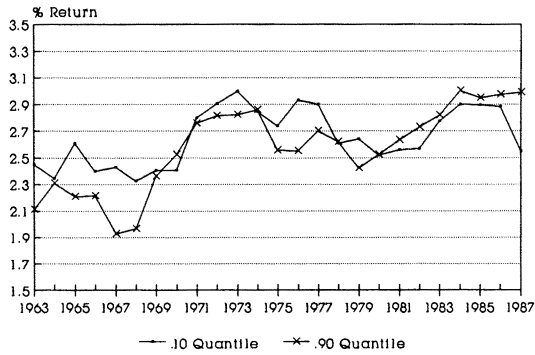
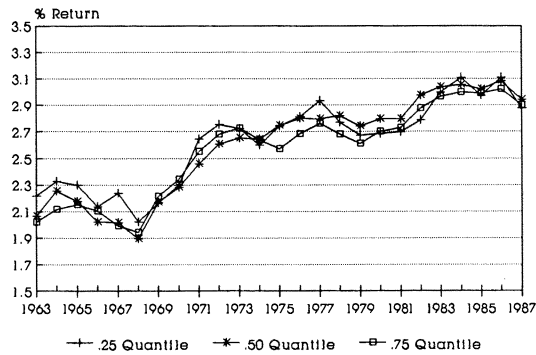
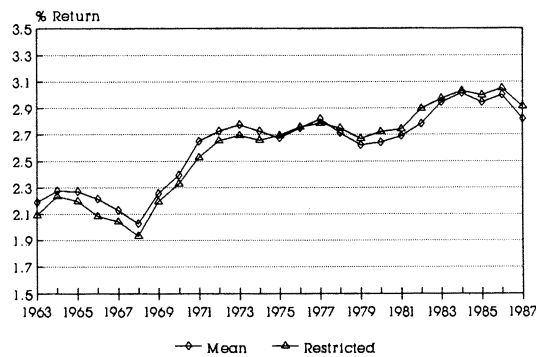
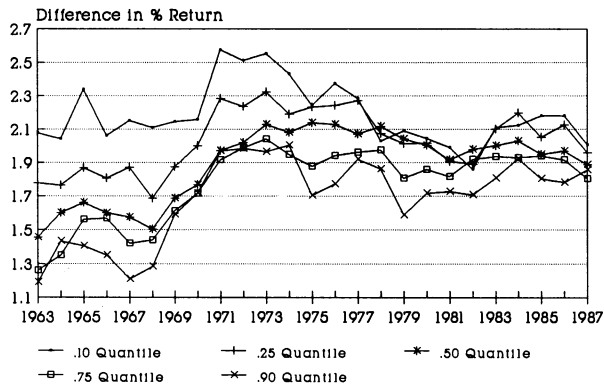
a. .10 and .90 Quantiles**b. .25, .50, and .75 Quantiles****c. Restricted across Quantiles and Mean**

FIGURE 3.—One-group model—return to experience for experienced workers (15 years of experience), by quantiles.

lower quantiles, while for experienced workers returns at the five quantiles have no particular order. Moreover, at each quantile, changes in the returns are smoother and more gradual for the experienced workers than for the new entrants. For both groups differences among the returns at the five quantiles are more pronounced during the early part of the sample period. The mean and the restricted returns changed in a similar pattern, with the mean return higher during the early part of the sample period and the restricted returns higher toward the end of the period.

As with the returns to education, the year-to-year fluctuations in the returns to experience are greater and have larger standard errors at the extreme quantiles than at the middle quantiles. Also, the standard errors for the experienced workers are about half those for the new entrants.

a. All Quantiles



b. Mean and Restricted Estimates

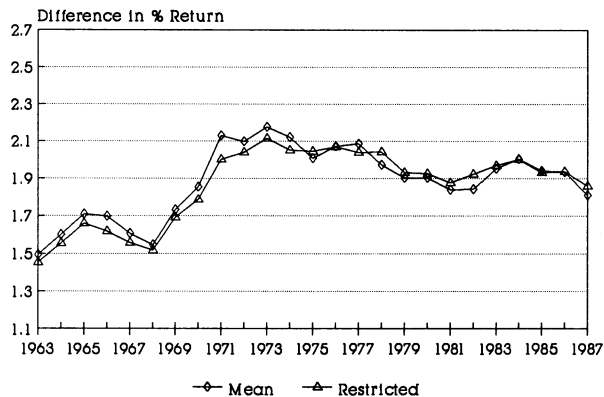


FIGURE 4.—One-group model—differences between the return to experience for new entrants and experienced workers, by quantiles.

The dissimilarities in the relationships of the returns across the five quantiles between the two experience groups are a consequence of the relationships between the coefficients of the linear and the quadratic terms of experience at the different quantiles. The coefficients of the linear terms are larger than those of the quadratic terms (which are all negative) at all quantiles, but the curvature of the return profile is more pronounced at the lower quantiles. The coefficients of both experience terms exhibit similar patterns but different magnitudes of changes at the different quantiles. Consequently, the returns to experience exhibit different behavior across the five quantiles for the two experience groups as is apparent from Table II and Figures 2 and 3.

The effect of smaller curvature in the experience profile at the higher quantiles is illustrated in Figure 4. Differences in the returns between the new entrants and experienced workers become smaller as the quantile increases, especially until the mid 1970's. Convergence of the returns for all quantiles toward the end of the period resulted in more comparable differences between the two groups.

4.1.3. *Summary*

The above results suggest that we do not have a location model. The five quantile regressions yield coefficient vector estimates which differ in all slope coefficients as well as in the constant term. Formal tests performed in each of the sample years strongly reject the null hypothesis of equality among the slope coefficients of the five vectors estimated.²⁹ Nevertheless, the similarities in the patterns of changes of the returns to education and experience, across the five quantiles, are remarkable.

The absence of interaction terms among the covariates in this simple model constrains the returns to change in the same manner at all skill levels. The model in the next section relaxes this constraint and allows the coefficients to change across the various education and experience levels.

A final observation worth noting relates to the coefficients on the dummy variable for blacks, which are depicted in Figure 5. The coefficients at the different quantiles differ widely during the first part of the sample period, but converge as time progresses. In 1964, for example, the wage for a black worker was smaller than for a white worker by a larger margin at the .10 quantile (52%) than at the .90 quantile (22%), holding everything else constant. The average black worker earned about 30% less than his white counterpart. Toward the end of the sample period, the wage for blacks was lower than for whites by about the same amount at all quantiles (31.1%, 25.1%, and 26.6% at the .10 and .90

²⁹ The null hypothesis of equality among all the slope coefficients (see column 9 of Table VI) is rejected for all years except 1984 at the 1% significance level; for 1984 rejection is at the 5% significance level. If the estimates were not statistically different they could be optimally combined, as is done in obtaining the restricted estimate. Efficiency can be increased by adding more quantile estimates. See the discussion on adaptive *l*-estimators in Portnoy and Koenker (1989).

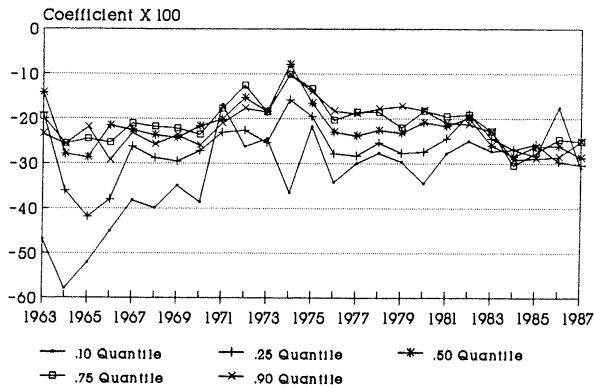
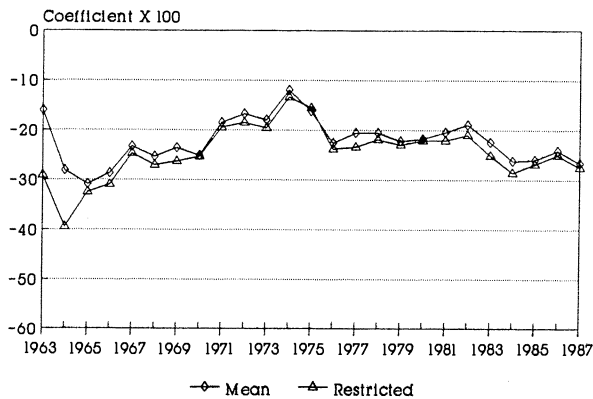
a. All Quantiles**b. Restricted and Mean Estimates**

FIGURE 5.—One-group model—coefficient on black dummy, by quantiles.

quantiles and the mean, respectively). Not only did the wage differential between blacks and whites decrease but it also became more uniform across the wage distribution.

4.2. 16-Group Model—Estimation by Groups

The model in this section goes two steps beyond the simple one-group model:

a. The sample is divided into 16 education-experience groups determined by the combination of four education groups: 0 to 7, 8 to 11, 12 to 15, and 16 + years of education; and four experience groups: 0 to 9, 10 to 19, 20 to 29, and 30 + years of experience.

b. Each regression includes additional variables: a part-time dummy, a metropolitan area dummy, a set of educational dummies, and a set of regional dummies.³⁰

A different estimation strategy is needed, both because there are many more variables, and also because censoring can be a severe problem for the highly skilled groups which tend to have most of the top coded wage data. For these two reasons I use the ILPA algorithm, which reduces to a simple LP algorithm when there is no censoring.

The model was estimated for each of the 16 skill groups at the same five quantiles as before, in each of the 25 years of the sample, for a total of 2000 regressions. Censoring caused problems (i.e., the estimated conditional quantiles for some of the observations are above the censoring point) in only 31 cases and always for the most highly educated group. In 1977, for example, the fraction of observations with the conditional quantiles below the censoring point at the .90 quantile, were .97, .89, and .91 for the groups with 10–19, 20–29, and 30 + years of experience (and 16 + years of education). In the next three years these fractions were .88, .85, and .71 in 1978; .82, .62, and .53 in 1979; and .66, .61, and .46 in 1980.

The estimated conditional quantiles for key years are reported in Table III along with their standard errors.³¹ Each conditional quantile is evaluated at a weighted average of all possible combinations of covariate values within each education-experience group.³² In general, the resulting values are consistent with the conditional quantiles increasing at all quantiles, in both experience and education. The only group for which the linear model seems not to fit perfectly is the 0–7 education group.³³

Several results are apparent from Table III: (a) the higher the skill level, the higher the conditional quantile; (b) within each experience level the higher the education level, the larger the increase (or the smaller the decrease) in the conditional quantiles, at all quantiles throughout the sample period; (c) the higher the quantile, the larger the increase in the conditional quantile during the sample period; and (d) the higher the skill level, and especially at the higher

³⁰ The part-time dummy takes the value 1 if a worker is working less than 35 hours per week and 0 otherwise. The metropolitan area dummy takes the value 1 if an individual is living in a metropolitan area, and 0 otherwise. Metropolitan areas are taken from the SMSA definition used by the CPS. There are 19 educational dummies, one for each of the 19 possible education levels: the j th dummy takes the value 1 if the level of education equals j , and 0 otherwise ($j = 0, \dots, 18$). There are four regional dummies, for the Northeast, North Central, South, and West regions (according to the CPS definition); each takes the value 1 if an individual is in the region and 0 otherwise.

³¹ The asymptotic covariance matrix of the estimated coefficient vector is computed using the bootstrap method. The order statistic method, which is less time consuming and easier to implement, gives standard errors which are smaller by 30% to 50% than the bootstrap standard errors.

³² There are 32 such possible values of covariates in the subvector that excludes the experience variable and the educational dummies. The weights within the specific experience group are the proportions in the sample. The educational dummies subvector includes the dummies weighted by their proportion in the specific educational group. The exact levels at which the experience is evaluated are indicated in the table.

³³ See Card and Krueger (1992a) for similar results.

TABLE III
16-GROUP MODEL—CONDITIONAL QUANTILES OF LOG WEEKLY INCOME
FROM WAGES AND SALARIES

Education	0-7	8-11	12-15	16 +	0-7	8-11	12-15	16 +	0-7	8-11	12-15	16 +
Year	Panel A: 10 Quantile											
	5 Years Experience				15 Years Experience				25 Years Experience			
1963	4.23 (0.04)	4.70 (0.06)	4.83 (0.04)	5.20 (0.07)	4.71 (0.08)	4.92 (0.04)	5.17 (0.04)	5.38 (0.09)	4.76 (0.07)	4.98 (0.05)	5.18 (0.04)	5.29 (0.11)
1964	4.73 (0.08)	4.75 (0.05)	4.85 (0.04)	5.29 (0.08)	4.81 (0.08)	5.10 (0.06)	5.26 (0.04)	5.43 (0.08)	4.75 (0.05)	5.06 (0.03)	5.25 (0.03)	5.57 (0.09)
1968	4.65 (0.03)	4.82 (0.04)	5.06 (0.03)	5.50 (0.05)	4.87 (0.03)	5.22 (0.03)	5.43 (0.02)	5.86 (0.03)	5.06 (0.04)	5.22 (0.03)	5.44 (0.02)	5.63 (0.07)
1969	4.50 (0.03)	5.00 (0.04)	5.15 (0.03)	5.46 (0.06)	4.91 (0.04)	5.26 (0.04)	5.52 (0.02)	5.74 (0.04)	4.98 (0.04)	5.20 (0.03)	5.53 (0.03)	5.77 (0.06)
1974	4.50 (0.07)	4.87 (0.04)	5.02 (0.02)	5.31 (0.05)	4.99 (0.05)	5.23 (0.04)	5.50 (0.02)	5.52 (0.06)	5.02 (0.07)	5.31 (0.06)	5.50 (0.04)	5.53 (0.05)
1975	4.54 (0.03)	4.86 (0.04)	5.08 (0.02)	5.44 (0.04)	4.94 (0.04)	5.18 (0.04)	5.37 (0.03)	5.60 (0.05)	5.03 (0.04)	5.17 (0.05)	5.40 (0.04)	5.63 (0.08)
1978	4.58 (0.06)	4.80 (0.04)	5.12 (0.02)	5.43 (0.03)	5.01 (0.03)	5.09 (0.04)	5.46 (0.03)	5.75 (0.03)	4.90 (0.07)	5.28 (0.05)	5.48 (0.03)	5.80 (0.05)
1979	4.77 (0.05)	4.88 (0.04)	5.07 (0.02)	5.45 (0.03)	4.97 (0.04)	5.18 (0.04)	5.49 (0.02)	5.75 (0.04)	4.93 (0.06)	5.40 (0.05)	5.52 (0.03)	5.81 (0.06)
1983	4.58 (0.06)	4.67 (0.03)	4.85 (0.02)	5.29 (0.04)	4.89 (0.05)	5.01 (0.05)	5.25 (0.02)	5.62 (0.03)	4.96 (0.08)	5.11 (0.05)	5.31 (0.03)	5.79 (0.05)
1986	4.36 (0.03)	4.61 (0.04)	4.85 (0.02)	5.42 (0.05)	4.78 (0.04)	4.99 (0.05)	5.31 (0.02)	5.69 (0.04)	4.93 (0.09)	5.13 (0.06)	5.39 (0.03)	5.84 (0.06)
1987	4.64 (0.06)	4.73 (0.05)	4.88 (0.02)	5.46 (0.03)	4.73 (0.05)	4.99 (0.05)	5.25 (0.03)	5.68 (0.04)	4.82 (0.05)	5.10 (0.08)	5.44 (0.03)	5.69 (0.05)

TABLE III—Continued

Education		0-7	8-11	12-15	16 +	0-7	8-11	12-15	16 +	0-7	8-11	12-15	16 +
Year		Panel B: 25 Quintile											
		5 Years Experience				15 Years Experience				25 Years Experience			
1963		4.74 (0.10)	4.93 (0.03)	5.17 (0.03)	5.49 (0.05)	5.00 (0.04)	5.23 (0.04)	5.50 (0.02)	5.73 (0.05)	5.08 (0.04)	5.32 (0.02)	5.43 (0.02)	5.62 (0.08)
1964		4.94 (0.07)	5.04 (0.03)	5.21 (0.03)	5.54 (0.04)	5.07 (0.03)	5.32 (0.02)	5.50 (0.02)	5.81 (0.05)	5.13 (0.04)	5.31 (0.03)	5.59 (0.02)	5.86 (0.08)
1968		5.07 (0.05)	5.14 (0.03)	5.40 (0.02)	5.76 (0.03)	5.12 (0.04)	5.43 (0.02)	5.70 (0.02)	6.04 (0.03)	5.36 (0.03)	5.51 (0.02)	5.71 (0.02)	5.98 (0.04)
1969		4.81 (0.06)	5.27 (0.02)	5.46 (0.02)	5.76 (0.03)	5.27 (0.03)	5.51 (0.03)	5.74 (0.02)	6.06 (0.03)	5.31 (0.03)	5.55 (0.02)	5.75 (0.01)	6.02 (0.04)
1974		4.94 (0.05)	5.22 (0.04)	5.33 (0.02)	5.61 (0.02)	5.29 (0.05)	5.59 (0.03)	5.81 (0.02)	6.02 (0.03)	5.38 (0.03)	5.59 (0.03)	5.83 (0.02)	6.10 (0.04)
1975		4.98 (0.04)	5.19 (0.04)	5.35 (0.02)	5.65 (0.01)	5.17 (0.03)	5.43 (0.04)	5.68 (0.02)	5.97 (0.04)	5.28 (0.04)	5.51 (0.04)	5.71 (0.02)	6.16 (0.05)
1978		5.02 (0.07)	5.17 (0.03)	5.41 (0.01)	5.71 (0.02)	5.17 (0.03)	5.45 (0.04)	5.76 (0.02)	6.07 (0.03)	5.24 (0.05)	5.64 (0.04)	5.82 (0.02)	6.14 (0.03)
1979		4.99 (0.04)	5.25 (0.03)	5.40 (0.01)	5.69 (0.02)	5.17 (0.04)	5.46 (0.04)	5.79 (0.02)	6.09 (0.02)	5.21 (0.06)	5.66 (0.03)	5.82 (0.02)	6.13 (0.02)
1983		4.88 (0.07)	4.94 (0.03)	5.18 (0.03)	5.60 (0.02)	5.07 (0.04)	5.39 (0.04)	5.59 (0.02)	5.94 (0.02)	5.21 (0.08)	5.46 (0.04)	5.71 (0.02)	6.10 (0.03)
1986		4.70 (0.05)	4.89 (0.03)	5.19 (0.02)	5.74 (0.03)	5.02 (0.03)	5.31 (0.05)	5.66 (0.02)	6.06 (0.02)	5.23 (0.07)	5.52 (0.05)	5.78 (0.02)	6.21 (0.03)
1987		4.74 (0.03)	5.02 (0.03)	5.19 (0.02)	5.79 (0.02)	4.98 (0.04)	5.32 (0.05)	5.63 (0.02)	6.00 (0.02)	5.10 (0.04)	5.48 (0.05)	5.80 (0.02)	6.13 (0.03)

TABLE III—Continued

Education	Panel C: .50 Quantile					25 Years Experience				
	0-7	8-11	12-15	16+	16+	0-7	8-11	12-15	16+	16+
Year	5 Years Experience					15 Years Experience				
1963	5.02 (0.08)	5.25 (0.04)	5.42 (0.02)	5.67 (0.03)	5.36 (0.05)	5.52 (0.02)	5.74 (0.02)	5.42 (0.03)	5.61 (0.02)	5.76 (0.02)
1964	5.15 (0.06)	5.30 (0.04)	5.46 (0.02)	5.76 (0.04)	5.28 (0.03)	5.55 (0.02)	5.76 (0.02)	5.43 (0.04)	5.63 (0.02)	5.82 (0.02)
1968	5.30 (0.05)	5.45 (0.03)	5.69 (0.01)	6.01 (0.02)	5.44 (0.02)	5.72 (0.02)	5.94 (0.01)	5.62 (0.02)	5.76 (0.02)	5.96 (0.02)
1969	5.22 (0.05)	5.57 (0.03)	5.70 (0.01)	6.06 (0.03)	5.58 (0.04)	5.79 (0.02)	5.97 (0.01)	5.60 (0.03)	5.83 (0.02)	5.99 (0.01)
1974	5.18 (0.06)	5.56 (0.04)	5.65 (0.01)	5.94 (0.02)	5.57 (0.06)	5.89 (0.03)	6.07 (0.01)	5.73 (0.03)	5.85 (0.03)	6.07 (0.02)
1975	5.28 (0.07)	5.49 (0.03)	5.61 (0.01)	5.90 (0.02)	5.44 (0.05)	5.70 (0.02)	5.98 (0.01)	5.63 (0.04)	5.82 (0.02)	6.02 (0.02)
1978	5.23 (0.05)	5.48 (0.03)	5.69 (0.01)	5.97 (0.02)	5.52 (0.03)	5.79 (0.04)	6.08 (0.01)	5.61 (0.05)	5.97 (0.03)	6.12 (0.01)
1979	5.24 (0.03)	5.52 (0.02)	5.70 (0.01)	5.95 (0.01)	5.55 (0.06)	5.78 (0.03)	6.09 (0.01)	5.60 (0.06)	5.95 (0.03)	6.13 (0.02)
1983	5.13 (0.04)	5.27 (0.03)	5.52 (0.01)	5.91 (0.02)	5.42 (0.04)	5.68 (0.04)	5.94 (0.01)	5.57 (0.03)	5.80 (0.04)	6.08 (0.02)
1986	4.94 (0.04)	5.21 (0.03)	5.52 (0.01)	6.05 (0.02)	5.24 (0.05)	5.65 (0.04)	5.98 (0.01)	5.49 (0.04)	5.85 (0.05)	6.14 (0.02)
1987	5.03 (0.05)	5.28 (0.03)	5.51 (0.01)	6.05 (0.02)	5.31 (0.03)	5.72 (0.03)	5.94 (0.02)	5.47 (0.05)	5.75 (0.05)	6.12 (0.02)

TABLE III—Continued

Education	0-7	8-11	12-15	16 +	0-7	8-11	12-15	16 +	0-7	8-11	12-15	16 +
Year	Panel D: .75 Quantile											
	5 Years Experience				15 Years Experience				25 Years Experience			
1963	5.29 (0.04)	5.51 (0.03)	5.65 (0.02)	5.90 (0.03)	5.58 (0.05)	5.74 (0.03)	5.94 (0.02)	6.22 (0.06)	5.66 (0.04)	5.86 (0.03)	5.98 (0.03)	6.26 (0.09)
1964	5.42 (0.04)	5.55 (0.03)	5.73 (0.02)	6.02 (0.05)	5.61 (0.03)	5.79 (0.03)	6.02 (0.02)	6.36 (0.04)	5.63 (0.03)	5.89 (0.02)	6.05 (0.02)	6.43 (0.05)
1968	5.50 (0.06)	5.74 (0.03)	5.92 (0.01)	6.24 (0.03)	5.75 (0.03)	5.95 (0.02)	6.15 (0.02)	6.49 (0.03)	5.87 (0.03)	6.01 (0.01)	6.19 (0.01)	6.63 (0.05)
1969	5.50 (0.04)	5.80 (0.03)	5.94 (0.01)	6.29 (0.02)	5.86 (0.04)	6.02 (0.02)	6.19 (0.01)	6.52 (0.03)	5.89 (0.02)	6.06 (0.02)	6.21 (0.01)	6.59 (0.05)
1974	5.68 (0.06)	5.84 (0.04)	5.92 (0.01)	6.21 (0.02)	5.83 (0.05)	6.16 (0.03)	6.28 (0.02)	6.58 (0.03)	6.04 (0.04)	6.13 (0.03)	6.31 (0.02)	6.71 (0.04)
1975	5.53 (0.02)	5.72 (0.04)	5.89 (0.01)	6.14 (0.02)	5.76 (0.05)	5.99 (0.02)	6.23 (0.01)	6.61 (0.02)	5.91 (0.04)	6.10 (0.03)	6.30 (0.02)	6.74 (0.04)
1978	5.50 (0.04)	5.80 (0.02)	5.99 (0.01)	6.25 (0.02)	5.76 (0.03)	6.14 (0.03)	6.33 (0.01)	6.69 (0.03)	5.99 (0.05)	6.21 (0.03)	6.41 (0.02)	6.73 (0.03)
1979	5.48 (0.04)	5.86 (0.03)	5.97 (0.01)	6.22 (0.02)	5.81 (0.05)	6.16 (0.03)	6.34 (0.01)	6.69 (0.02)	5.93 (0.04)	6.20 (0.03)	6.38 (0.02)	6.77 (0.03)
1983	5.40 (0.05)	5.56 (0.03)	5.82 (0.01)	6.20 (0.02)	5.66 (0.03)	6.01 (0.03)	6.23 (0.01)	6.59 (0.02)	5.94 (0.05)	6.19 (0.04)	6.38 (0.02)	6.74 (0.03)
1986	5.16 (0.05)	5.50 (0.04)	5.84 (0.02)	6.29 (0.02)	5.63 (0.07)	5.94 (0.05)	6.27 (0.01)	6.70 (0.02)	5.90 (0.04)	6.18 (0.04)	6.40 (0.02)	6.91 (0.03)
1987	5.31 (0.04)	5.55 (0.03)	5.81 (0.01)	6.33 (0.03)	5.60 (0.08)	6.05 (0.03)	6.23 (0.01)	6.65 (0.02)	5.89 (0.07)	6.07 (0.04)	6.39 (0.02)	6.83 (0.03)

TABLE III—Continued

Education	Panel E: .90 Quantile					25 Years Experience				
	0-7	8-11	12-15	16 +		0-7	8-11	12-15	16 +	
Year	5 Years Experience					15 Years Experience				
1963	5.74 (0.04)	5.73 (0.04)	5.86 (0.03)	6.13 (0.06)		5.83 (0.07)	5.95 (0.03)	6.12 (0.03)	6.53 (0.08)	5.98 (0.04)
1964	5.60 (0.09)	5.81 (0.05)	5.95 (0.03)	6.26 (0.05)		5.81 (0.03)	6.07 (0.04)	6.20 (0.03)	6.54 (0.05)	6.01 (0.04)
1968	5.97 (0.05)	5.99 (0.03)	6.15 (0.03)	6.38 (0.03)		5.98 (0.03)	6.18 (0.03)	6.34 (0.02)	6.74 (0.03)	6.10 (0.03)
1969	5.86 (0.03)	6.00 (0.03)	6.16 (0.02)	6.52 (0.04)		6.01 (0.04)	6.25 (0.03)	6.36 (0.02)	6.72 (0.04)	6.10 (0.02)
1974	6.00 (0.03)	6.07 (0.05)	6.13 (0.02)	6.44 (0.02)		5.98 (0.04)	6.40 (0.03)	6.51 (0.02)	6.88 (0.06)	6.31 (0.03)
1975	5.68 (0.02)	5.95 (0.05)	6.10 (0.02)	6.38 (0.03)		6.02 (0.04)	6.19 (0.03)	6.44 (0.01)	6.89 (0.05)	6.11 (0.03)
1978	5.84 (0.04)	6.07 (0.03)	6.24 (0.01)	6.47 (0.02)		6.10 (0.02)	6.36 (0.03)	6.54 (0.02)	6.99 (0.03)	6.17 (0.02)
1979	5.75 (0.02)	6.16 (0.05)	6.24 (0.01)	6.44 (0.02)		6.06 (0.05)	6.37 (0.03)	6.55 (0.01)	6.99 (0.04)	6.19 (0.06)
1983	5.79 (0.02)	5.83 (0.06)	6.12 (0.02)	6.44 (0.02)		6.07 (0.10)	6.28 (0.05)	6.47 (0.02)	6.88 (0.03)	6.26 (0.03)
1986	5.41 (0.07)	5.79 (0.04)	6.12 (0.02)	6.55 (0.03)		6.04 (0.04)	6.21 (0.04)	6.50 (0.02)	7.00 (0.04)	6.26 (0.06)
1987	5.52 (0.02)	5.82 (0.04)	6.05 (0.02)	6.55 (0.02)		6.11 (0.04)	6.29 (0.04)	6.52 (0.02)	6.97 (0.03)	6.24 (0.05)

Note: The regressions are computed using the iterative linear programming algorithm described in Section 3. The regressions are estimated for 16 different education-experience groups generated by four education groups (0-7, 8-11, 12-15, and 16+) and four experience groups (0-9, 10-19, 20-29, and 30+). The regressions include a full set of educational dummies, experience, experience squared, a dummy variable for blacks, four regional dummies, a part time dummy, and a metropolitan area dummy. The reported numbers are of the estimated conditional log weekly wage evaluated according to the description in footnote 32. The sample is obtained exactly according to the selection criteria outlined in Section 2. The standard errors are computed using the bootstrap method described in Section 3.

level of education, the smaller the year-to-year fluctuations in the conditional quantiles.

4.2.1. 16-Group Model—Return to Education

Having solved the model for each skill group separately with a full set of educational dummies, we can then measure the return to education as the relative increase in the conditional quantile between two adjacent education groups with the same level of experience, holding the other variables constant. For example, the return to college education for workers with 25 years of experience at the θ th quantile is defined in percentages by

$$R_{a_{16},25}^{ED} = 100 \times \left(\frac{Q_{a_{16},25}^{\theta} - Q_{a_{12},25}^{\theta}}{a_{16} - a_{12}} \right),$$

where a_{12} and a_{16} are the weighted averages of the education dummy variables for the 12–15 years and 16+ years of education, respectively, and $Q_{l,k}^{\theta}$ is the θ th conditional quantile evaluated at the weighted average of the education dummy variable l and k years of experience. Since the returns are computed from Table III, they are representative for the whole population within each skill group.³⁴ The returns to education for key years are reported in Table IV and graphed in Figures 6, 7, and 8 for the new entrants, experienced workers, and prime earners, respectively. Because of the way in which the returns are computed, the title: “Return to Education for High School Graduates” (in Figure 6b, for example) should be understood as: “the return of high school graduation relative to the 8–11 years of education group,” and similarly for the other groups.

There is an interesting contrast between these results and those previously presented for the one-group model. The latter showed a significant general increase during the 1980’s in the return to education across all quantiles. The present results indicate that this overall increase is due primarily to the increase in the return to: (a) high school and, especially, college education for the new entrants; and (b) college education for the experienced workers.

While the overall picture revealed by this model is not as transparent, several general points are clear: (a) the return to college education is higher, in general, than for high school education at every quantile and for all experience groups; (b) the returns at the five quantiles are ranked in a different order for the different skill groups; and (c) exceedingly high year-to-year fluctuations in the returns are apparent for the high school dropouts, at all quantiles, with no clear pattern of change.

The returns to education for the college graduates new entrants group are higher, in general, at the lower quantiles, and higher at the higher quantiles for

³⁴ I also considered an alternative calculation for the returns to education which is based on differences between conditional quantiles evaluated at 16, 12, 8, and 0 years of education. The returns reported in the paper are slightly smoother (across the years) than those based on the alternative method. Differences between the two methods are, nonetheless, fairly minor.

TABLE IV
16-GROUP MODEL—RETURN TO EDUCATION BY QUANTILES
DEPENDENT VARIABLE: LOG OF WEEKLY INCOME FROM WAGES AND SALARIES

Quantile	.10	.25	.50	.75	.90	.10	.25	.50	.75	.90	.10	.25	.50	.75	.90
Panel A: 5 Years of Experience															
Year	8 Years of Education					12 Years of Education					16 Years of Education				
1963	10.6 (1.6)	9.2 (2.4)	11.7 (2.0)	11.0 (1.1)	-0.7 (1.3)	4.4 (2.4)	5.9 (1.4)	4.2 (1.5)	3.7 (1.4)	3.2 (1.8)	8.9 (1.9)	8.0 (1.4)	6.4 (1.0)	6.2 (0.9)	6.7 (1.7)
1964	0.7 (2.5)	4.9 (1.9)	7.8 (1.7)	6.3 (1.3)	10.8 (2.5)	3.5 (2.3)	4.2 (1.5)	4.0 (1.6)	4.5 (1.5)	3.3 (2.1)	10.6 (2.2)	8.3 (1.2)	7.4 (1.1)	7.5 (1.3)	7.8 (1.5)
1968	4.2 (1.1)	3.7 (1.3)	7.3 (1.3)	11.8 (1.5)	1.1 (1.5)	8.3 (1.6)	6.5 (1.1)	6.1 (1.0)	4.6 (1.1)	4.1 (1.4)	10.8 (1.4)	8.9 (0.9)	8.0 (0.5)	8.0 (0.8)	5.7 (1.0)
1969	11.5 (1.1)	23.3 (1.4)	17.4 (1.2)	15.1 (1.1)	6.7 (1.0)	5.3 (1.8)	4.7 (1.1)	3.4 (1.0)	3.3 (1.0)	4.1 (1.1)	7.7 (1.6)	7.5 (0.8)	8.9 (0.8)	8.9 (0.7)	9.0 (1.0)
1974	8.3 (1.8)	14.1 (1.5)	18.9 (1.7)	8.3 (1.7)	3.8 (1.3)	5.9 (1.7)	2.7 (1.7)	2.3 (1.5)	1.8 (1.4)	1.5 (2.0)	7.1 (1.3)	7.0 (0.7)	7.2 (0.6)	7.4 (0.6)	7.8 (0.7)
1975	6.8 (1.1)	10.1 (1.2)	10.4 (1.6)	9.5 (0.9)	13.7 (1.1)	8.3 (1.8)	4.2 (1.6)	3.0 (1.2)	4.1 (1.4)	3.6 (1.9)	9.0 (1.1)	7.5 (0.6)	7.2 (0.5)	6.3 (0.7)	7.2 (0.8)
1978	4.7 (1.5)	7.4 (1.5)	12.2 (1.1)	15.3 (1.1)	11.3 (1.1)	12.1 (1.6)	5.9 (1.1)	5.4 (1.1)	4.6 (1.0)	4.3 (1.2)	7.9 (0.8)	7.5 (0.6)	6.8 (0.5)	6.6 (0.5)	5.9 (0.6)
1979	2.2 (1.2)	12.7 (1.0)	14.1 (0.8)	18.8 (1.1)	20.9 (1.0)	7.0 (1.6)	3.8 (1.2)	4.4 (1.0)	2.7 (1.3)	1.9 (1.8)	9.6 (0.9)	7.3 (0.6)	6.4 (0.5)	6.3 (0.5)	5.1 (0.7)
1983	1.9 (1.3)	2.8 (1.6)	6.8 (0.9)	7.8 (1.2)	1.9 (1.2)	6.6 (1.4)	6.0 (1.3)	6.4 (1.2)	6.5 (1.3)	7.2 (2.2)	11.1 (1.0)	10.5 (0.7)	9.7 (0.6)	9.7 (0.6)	8.1 (0.7)
1986	5.1 (0.9)	9.4 (1.2)	13.8 (1.1)	17.2 (1.3)	18.9 (1.7)	8.6 (1.5)	7.5 (1.2)	7.8 (1.2)	8.5 (1.5)	8.1 (1.7)	14.8 (1.3)	13.7 (0.9)	13.1 (0.7)	11.2 (0.7)	10.9 (0.9)
1987	1.7 (1.5)	14.3 (0.8)	12.6 (1.2)	11.7 (1.0)	14.8 (0.9)	5.8 (2.0)	4.1 (1.4)	5.7 (1.3)	6.4 (1.4)	5.8 (1.7)	14.9 (1.0)	15.1 (0.8)	13.4 (0.7)	13.1 (0.8)	12.4 (0.8)

TABLE IV—Continued

Quantile	.10	.25	.50	.75	.90	.10	.25	.50	.75	.90	.10	.25	.50	.75	.90
Panel C: 25 Years of Experience															
16 Years of Education															
Year	8 Years of Education					12 Years of Education					16 Years of Education				
1963	5.1 (2.2)	11.7 (1.0)	10.0 (0.9)	10.1 (1.1)	4.4 (1.4)	6.4 (2.0)	2.8 (0.9)	3.6 (1.0)	3.1 (1.2)	4.7 (1.9)	2.8 (2.7)	4.9 (1.9)	5.9 (1.3)	6.9 (2.3)	4.8 (2.8)
1964	7.6 (1.4)	9.0 (1.3)	10.0 (1.0)	12.7 (0.9)	4.1 (1.5)	6.0 (1.5)	7.0 (1.2)	4.8 (1.0)	4.2 (1.0)	4.0 (1.9)	7.6 (2.3)	6.7 (2.0)	8.1 (1.4)	9.4 (1.3)	6.5 (2.1)
1968	3.8 (1.3)	7.1 (0.9)	7.2 (0.8)	6.9 (0.8)	5.9 (0.9)	7.3 (1.2)	5.2 (0.8)	5.1 (0.8)	4.5 (0.7)	5.8 (1.1)	4.3 (1.7)	6.7 (1.0)	7.9 (1.0)	11.2 (1.2)	10.4 (1.3)
1969	5.4 (1.3)	12.0 (0.9)	11.8 (0.8)	8.3 (0.7)	8.8 (0.8)	10.5 (1.3)	5.0 (0.7)	4.0 (0.7)	3.8 (0.7)	4.8 (0.9)	5.7 (1.7)	6.7 (1.1)	8.1 (1.0)	9.5 (1.2)	9.3 (2.2)
1974	6.9 (2.2)	10.1 (1.0)	6.0 (1.0)	4.8 (1.2)	3.6 (1.2)	6.0 (2.4)	6.2 (1.0)	5.5 (1.1)	4.6 (1.1)	5.1 (1.3)	0.7 (1.5)	6.7 (1.1)	9.7 (0.9)	9.9 (1.1)	9.3 (1.5)
1975	3.3 (1.5)	11.2 (1.2)	9.4 (1.1)	9.6 (1.0)	10.5 (1.1)	7.2 (2.0)	5.0 (1.3)	4.9 (0.9)	5.1 (1.0)	4.9 (1.3)	5.6 (2.1)	11.3 (1.2)	11.1 (0.9)	11.0 (1.1)	14.1 (2.0)
1978	8.7 (1.9)	19.7 (1.5)	17.8 (1.3)	11.1 (1.4)	13.3 (0.8)	6.7 (1.8)	4.6 (1.6)	4.0 (1.0)	4.8 (1.1)	4.8 (1.2)	7.3 (1.3)	8.0 (0.9)	8.5 (0.6)	8.2 (0.8)	10.2 (0.8)
1979	10.7 (1.7)	22.5 (1.5)	17.3 (1.5)	13.2 (1.1)	13.4 (1.5)	4.0 (1.8)	4.0 (1.2)	4.6 (1.1)	4.6 (1.0)	4.2 (1.0)	6.5 (1.4)	7.7 (0.7)	7.6 (0.7)	9.8 (0.8)	11.5 (0.6)
1983	3.4 (2.2)	12.4 (1.9)	11.7 (1.0)	12.3 (1.5)	7.2 (1.0)	6.6 (2.1)	6.3 (1.5)	7.0 (1.3)	4.7 (1.5)	5.3 (1.3)	10.9 (1.3)	9.7 (0.8)	8.6 (0.7)	9.0 (0.8)	15.2 (0.9)
1986	4.7 (2.5)	14.2 (2.0)	18.1 (1.5)	14.1 (1.1)	6.1 (1.5)	8.5 (2.2)	6.5 (1.8)	7.4 (1.6)	5.5 (1.3)	6.7 (1.5)	10.3 (1.6)	11.0 (0.9)	10.6 (0.8)	12.9 (0.8)	17.7 (1.3)
1987	5.8 (2.0)	19.1 (1.4)	13.8 (1.5)	9.1 (1.7)	4.1 (1.4)	11.2 (2.9)	8.2 (1.8)	9.3 (1.7)	8.1 (1.5)	8.2 (1.8)	5.6 (1.5)	8.2 (0.8)	9.2 (0.7)	10.9 (0.8)	13.3 (1.0)

Note: See note in Table III and the text in Section 4.2.1.

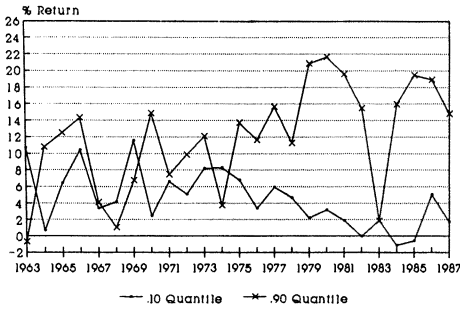
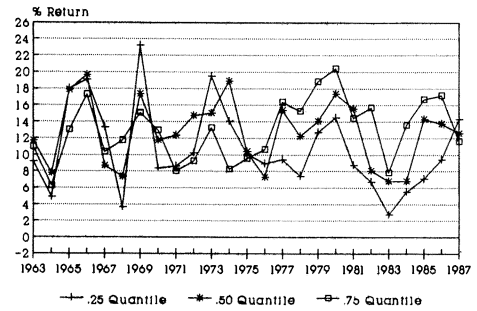
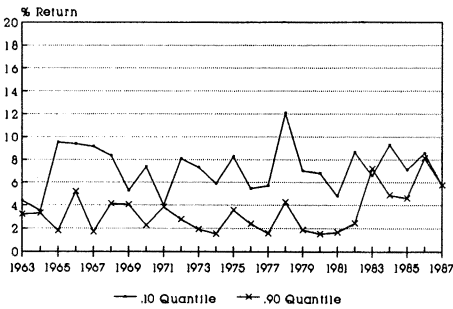
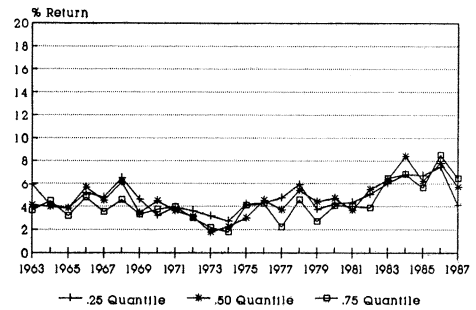
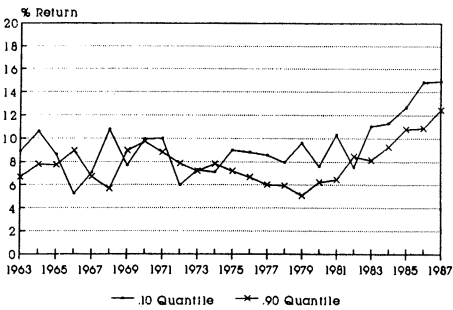
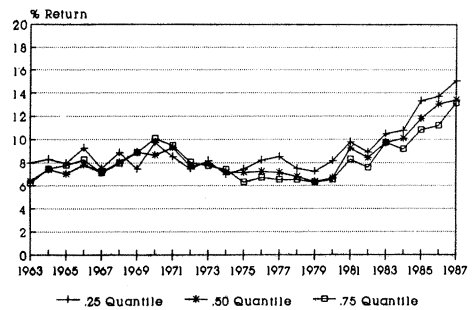
a. High School Dropouts**b. High School Dropouts****c. High School Graduates****d. High School Graduates****e. College Graduates****f. College Graduates**

FIGURE 6.—16-group model—return to education for new entrants (5 years of experience), by quantiles.

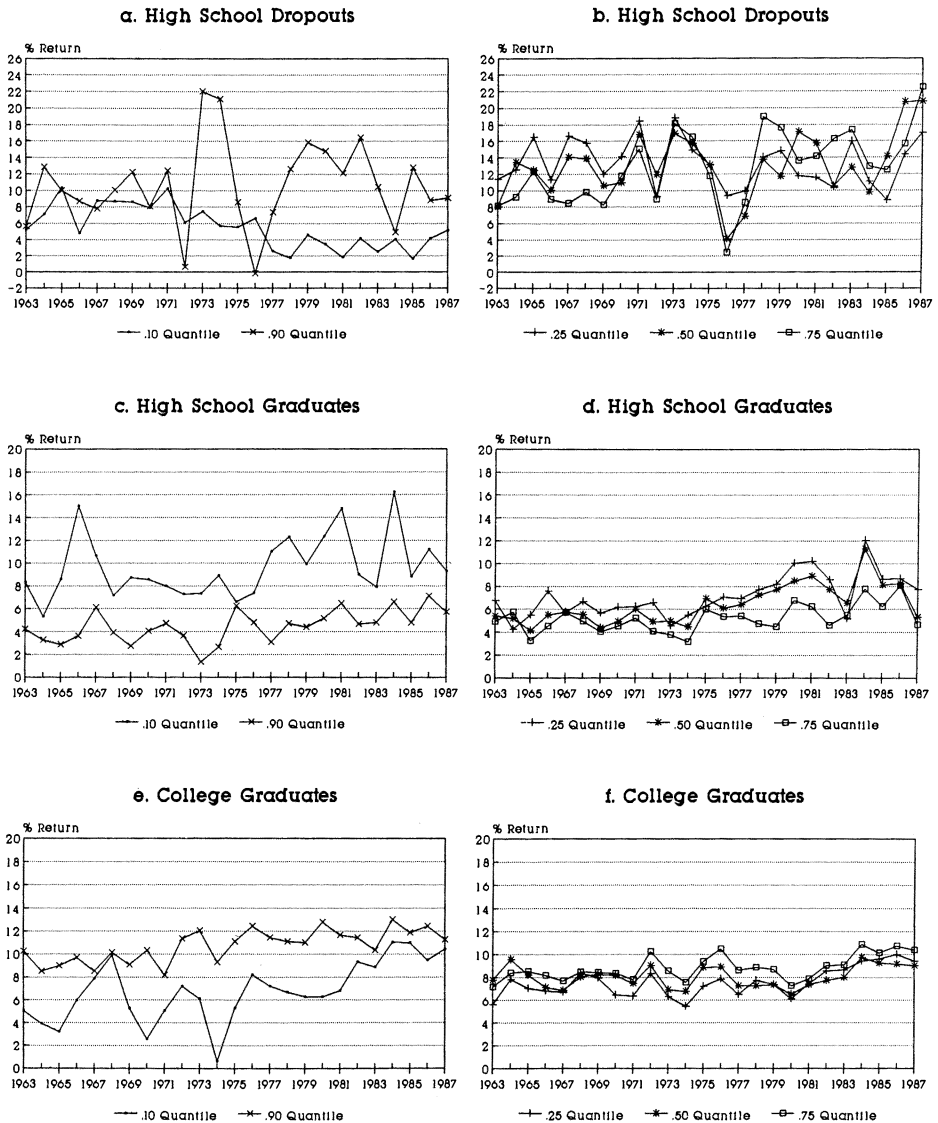


FIGURE 7.—16-group model—return to education for experienced workers (15 years of experience), by quantiles.

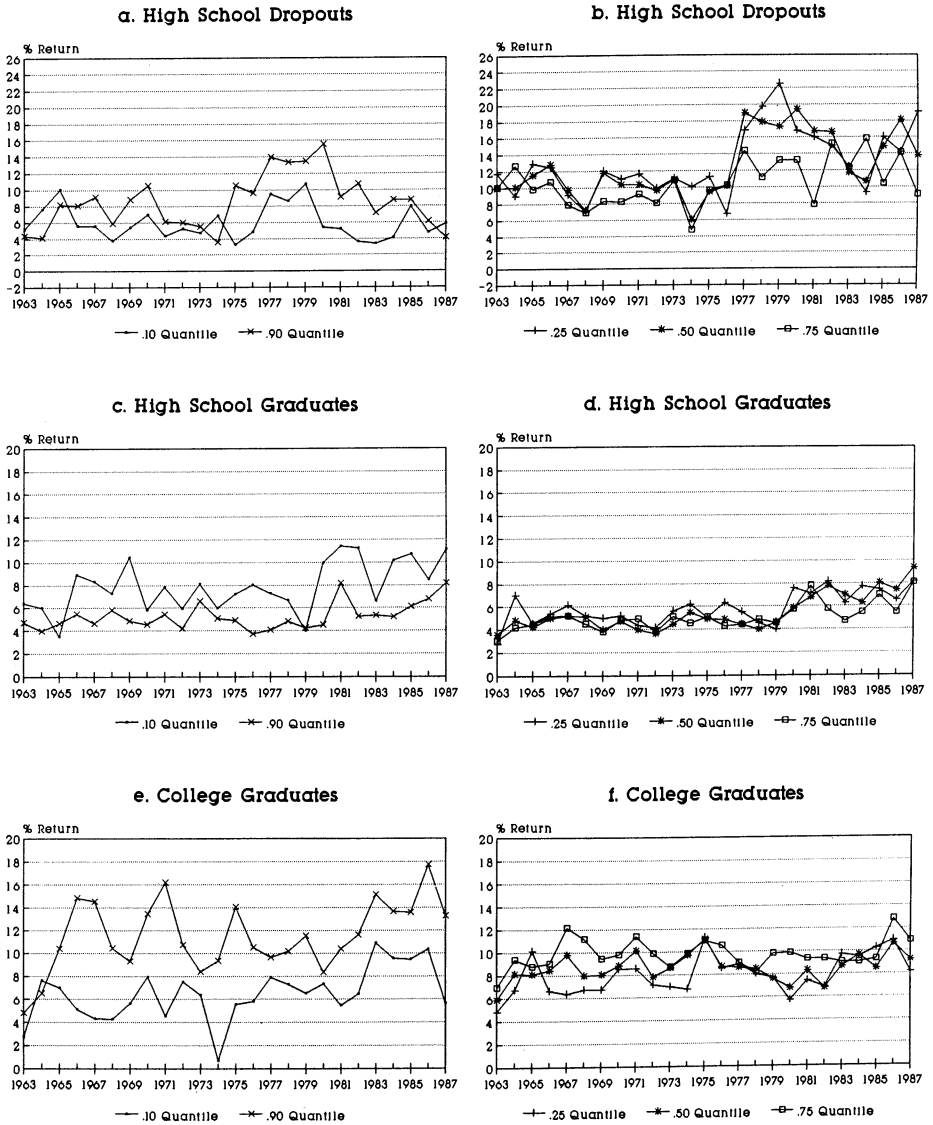


FIGURE 8.—16-group model—return to education for prime earners (25 years of experience), by quantiles.

the experienced and prime earner workers. For the high school graduates, the returns are higher at lower quantiles for all experience groups. No clear order in the returns across the five quantiles can be detected for the high school dropouts.

The increase in the return to college education is most noticeable for the new entrants group, but the experienced workers and prime earners experienced significant increases as well. The increase in the return to high school education was not as large, but the pattern of the changes across the five quantiles was similar for all experience groups. In light of these results, it is clear that the one-group model underestimates the increases in the returns to college education and overestimates the increases in the returns to high school education.

For the high school dropouts the year-to-year changes in the return to education were quite large at all experience levels, with the larger changes at the higher quantiles. The returns at the different quantiles are not arranged in any particular order, and do not show any clear pattern of change.

To summarize, the 16-group model reveals several important findings: (a) changes in the return to education vary across quantiles and across experience levels; (b) the return to education is greater for higher education at the upper end of the wage distribution; and (c) patterns of change are similar for high school and college education at the different quantiles, but not for the high school dropouts. Statistical tests, which for brevity are not reported here, reject null hypotheses of equality among the slope coefficients at the five quantiles, for all years and for all 16 skill levels. The results indicate that the one-group model is too restrictive in its functional form. A more flexible form in a single-equation model should yield results more compatible with the 16-group model.

4.2.2. 16-Group Model—Return to Experience

The returns to experience, which are defined exactly as for the one-group model, are allowed to vary across the 16 skill groups. The returns for the new entrants and experienced workers in key years are reported in Table V and graphed in Figures 9 and 10, respectively. (The returns for the prime earners group are omitted since they are quite small and changed very little in absolute terms over the sample period.)

Figures 9 and 10 show completely different patterns of change for the two experience groups. In particular: (a) differences among the returns at the five quantiles are larger for the new entrants group than for the experienced workers, during the 1960's and the 1970's; (b) in general, the returns are higher at the lower quantiles for the new entrants with high school and college education, and higher at the higher quantiles for the experienced workers with college education; (c) the returns to experience at the five quantiles converged toward the end of the sample period for both experience groups; and (d) the returns to experience changed less (in absolute terms) than the returns to education, at all quantiles.

TABLE V
16-GROUP MODEL—RETURN TO EXPERIENCE BY QUANTILES
DEPENDENT VARIABLE: LOG OF WEEKLY INCOME FROM WAGES AND SALARIES

Quantile	.10	.25	.50	.75	.90	.10	.25	.50	.75	.90	.10	.25	.50	.75	.90
Panel A: 5 Years of Experience															
Year	8 Years of Education					12 Years of Education					16 Years of Education				
1963	4.3 (0.9)	2.6 (2.3)	2.9 (1.9)	5.8 (0.9)	5.1 (1.0)	9.1 (1.3)	8.9 (0.7)	7.3 (0.8)	6.9 (0.7)	6.3 (0.9)	9.9 (0.6)	8.9 (0.5)	7.6 (0.4)	6.4 (0.4)	6.4 (0.6)
1964	0.7 (1.9)	3.5 (1.5)	6.6 (1.3)	6.2 (0.8)	5.6 (2.0)	9.5 (1.2)	7.9 (0.8)	7.6 (0.9)	5.9 (0.8)	5.4 (1.1)	9.7 (0.7)	8.9 (0.5)	7.1 (0.4)	6.6 (0.4)	6.1 (0.6)
1968	-0.5 (0.8)	3.0 (1.3)	5.4 (1.3)	5.6 (1.5)	8.0 (1.4)	9.8 (0.8)	8.2 (0.6)	7.2 (0.6)	6.4 (0.6)	4.6 (0.7)	11.0 (0.6)	8.9 (0.4)	7.6 (0.3)	6.5 (0.3)	5.1 (0.6)
1969	7.2 (0.7)	8.4 (1.4)	7.2 (1.2)	8.2 (1.0)	9.8 (0.9)	9.5 (0.9)	9.6 (0.6)	7.9 (0.6)	6.9 (0.6)	6.4 (0.6)	10.4 (0.7)	9.3 (0.4)	8.7 (0.3)	7.6 (0.3)	6.7 (0.4)
1974	6.8 (1.9)	5.9 (1.4)	4.4 (1.8)	3.7 (1.7)	5.5 (0.7)	9.8 (0.9)	8.5 (0.9)	8.6 (0.8)	7.4 (0.8)	7.9 (1.2)	10.1 (0.5)	9.4 (0.4)	8.9 (0.3)	7.7 (0.3)	7.1 (0.4)
1975	0.1 (0.7)	0.7 (1.1)	1.6 (1.8)	1.1 (0.6)	3.8 (0.5)	6.2 (1.0)	5.8 (0.9)	4.1 (0.7)	4.3 (0.8)	4.3 (1.2)	8.0 (0.6)	8.0 (0.4)	7.4 (0.3)	7.2 (0.3)	6.6 (0.4)
1978	-0.2 (1.5)	2.6 (1.5)	4.4 (1.1)	2.2 (1.1)	3.6 (0.5)	6.4 (1.0)	6.2 (0.7)	7.0 (0.7)	7.0 (0.6)	7.2 (0.7)	6.9 (0.5)	7.0 (0.3)	6.9 (0.3)	6.5 (0.3)	5.9 (0.4)
1979	2.6 (1.1)	2.9 (1.0)	3.7 (0.8)	3.6 (1.1)	3.9 (0.5)	7.3 (0.9)	6.5 (0.7)	6.3 (0.6)	6.7 (0.8)	6.2 (1.1)	6.9 (0.5)	7.0 (0.3)	7.1 (0.2)	6.6 (0.3)	5.9 (0.4)
1983	5.6 (1.5)	5.6 (1.9)	6.1 (1.0)	6.3 (1.3)	6.2 (0.6)	2.8 (0.8)	5.1 (0.7)	6.1 (0.7)	8.4 (0.7)	8.0 (1.3)	6.6 (0.4)	7.7 (0.4)	8.2 (0.3)	8.1 (0.3)	8.2 (0.4)
1986	-1.1 (0.7)	-0.2 (1.3)	3.1 (1.0)	3.7 (1.3)	9.1 (1.7)	7.3 (0.8)	6.9 (0.6)	7.1 (0.7)	7.2 (0.9)	8.1 (1.0)	7.9 (0.5)	7.6 (0.4)	7.8 (0.3)	8.2 (0.4)	7.5 (0.4)
1987	5.4 (1.5)	5.1 (0.8)	6.2 (1.4)	2.6 (1.1)	4.6 (0.5)	4.0 (1.1)	4.7 (0.7)	4.9 (0.7)	5.8 (0.7)	6.0 (0.9)	6.5 (0.5)	7.3 (0.4)	7.8 (0.3)	8.0 (0.3)	8.0 (0.5)

TABLE V—Continued

Quantile	.10	.25	.50	.75	.90	.10	.25	.50	.75	.90	.10	.25	.50	.75	.90
Panel B: 15 Years of Experience															
Year	8 Years of Education					12 Years of Education					16 Years of Education				
1963	2.7 (0.9)	1.5 (0.8)	1.1 (0.5)	1.7 (0.7)	2.0 (0.6)	1.9 (0.6)	1.7 (0.4)	1.6 (0.3)	1.1 (0.3)	1.9 (0.6)	1.3 (1.5)	1.2 (0.8)	0.7 (0.8)	2.1 (0.9)	1.0 (1.2)
1964	-0.6 (1.3)	0.4 (0.5)	1.2 (0.4)	0.8 (0.6)	0.6 (0.7)	0.5 (0.7)	0.5 (0.4)	1.0 (0.4)	0.8 (0.3)	0.0 (0.5)	1.9 (1.5)	1.8 (0.9)	1.8 (0.7)	3.2 (0.7)	1.4 (0.8)
1968	2.8 (0.7)	2.0 (0.5)	1.0 (0.4)	1.1 (0.4)	1.0 (0.7)	1.1 (0.4)	1.2 (0.4)	1.2 (0.3)	1.5 (0.3)	1.8 (0.4)	-0.4 (0.6)	0.1 (0.5)	1.1 (0.5)	2.1 (0.5)	3.1 (0.6)
1969	1.5 (0.9)	1.4 (0.6)	0.8 (0.5)	0.5 (0.4)	-0.3 (0.7)	0.8 (0.5)	1.0 (0.3)	1.1 (0.2)	1.1 (0.3)	1.8 (0.4)	2.5 (0.7)	1.1 (0.5)	1.0 (0.5)	2.1 (0.6)	3.8 (0.8)
1974	0.7 (0.9)	2.4 (0.8)	1.5 (0.6)	0.2 (0.7)	0.8 (0.8)	2.3 (0.6)	2.0 (0.4)	1.6 (0.3)	1.6 (0.4)	1.9 (0.4)	0.3 (1.0)	0.9 (0.6)	2.3 (0.5)	1.9 (0.5)	4.0 (1.1)
1975	2.1 (1.0)	1.6 (0.8)	2.1 (0.5)	1.8 (0.5)	1.9 (0.7)	1.4 (0.7)	1.2 (0.4)	1.6 (0.3)	1.7 (0.3)	1.9 (0.3)	3.6 (1.1)	3.0 (0.8)	2.7 (0.4)	1.8 (0.5)	4.7 (1.0)
1978	2.1 (0.8)	1.8 (0.8)	2.9 (0.9)	1.4 (0.7)	1.0 (0.7)	2.1 (0.7)	2.6 (0.4)	1.8 (0.3)	2.3 (0.3)	2.3 (0.5)	-0.8 (0.8)	-0.7 (0.6)	0.6 (0.6)	1.5 (0.7)	2.3 (0.7)
1979	3.2 (0.8)	1.1 (0.7)	1.6 (0.6)	0.7 (0.7)	0.3 (0.7)	2.8 (0.5)	2.5 (0.4)	2.2 (0.2)	2.0 (0.3)	1.9 (0.3)	1.9 (0.9)	1.5 (0.5)	2.1 (0.5)	2.7 (0.6)	3.6 (0.9)
1983	1.9 (1.1)	1.0 (0.9)	3.1 (0.9)	2.9 (0.7)	1.5 (1.0)	2.3 (0.5)	1.9 (0.4)	2.0 (0.3)	2.2 (0.3)	2.4 (0.4)	2.6 (0.8)	2.1 (0.5)	2.6 (0.5)	3.3 (0.4)	3.3 (0.6)
1986	3.0 (1.0)	2.6 (1.1)	2.1 (0.8)	2.4 (1.0)	1.0 (0.8)	2.7 (0.5)	2.5 (0.4)	1.9 (0.3)	1.9 (0.3)	2.3 (0.4)	1.2 (0.9)	2.0 (0.5)	1.5 (0.5)	1.5 (0.5)	1.5 (0.9)
1987	0.1 (1.1)	1.4 (1.1)	1.9 (0.8)	2.4 (0.8)	1.4 (0.9)	2.3 (0.6)	2.7 (0.4)	2.5 (0.4)	1.9 (0.3)	2.1 (0.4)	1.0 (0.8)	1.7 (0.5)	1.8 (0.4)	2.1 (0.5)	2.6 (0.7)

Note: See note in Table III and the text in Section 4.2.2.

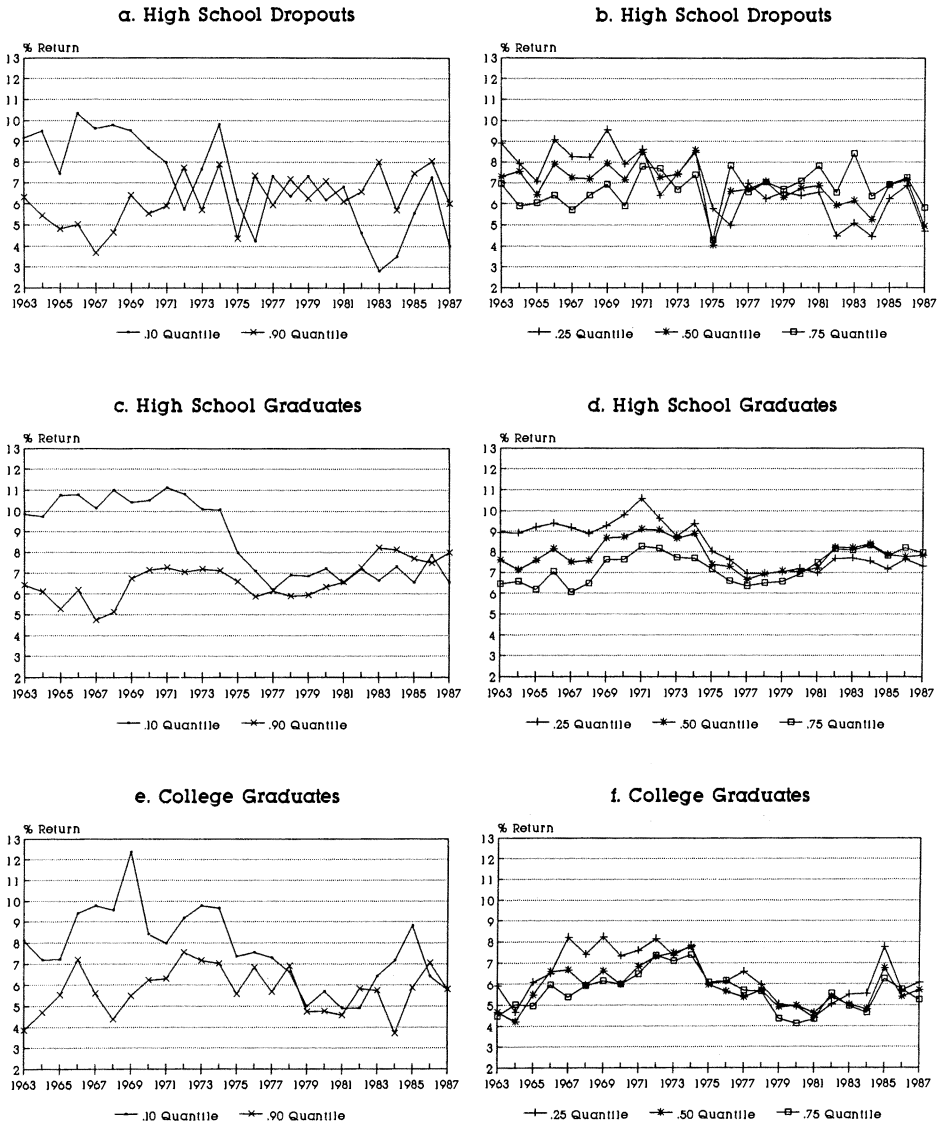


FIGURE 9.—16-group model—return to experience for new entrants (5 years of experience), by quantiles.

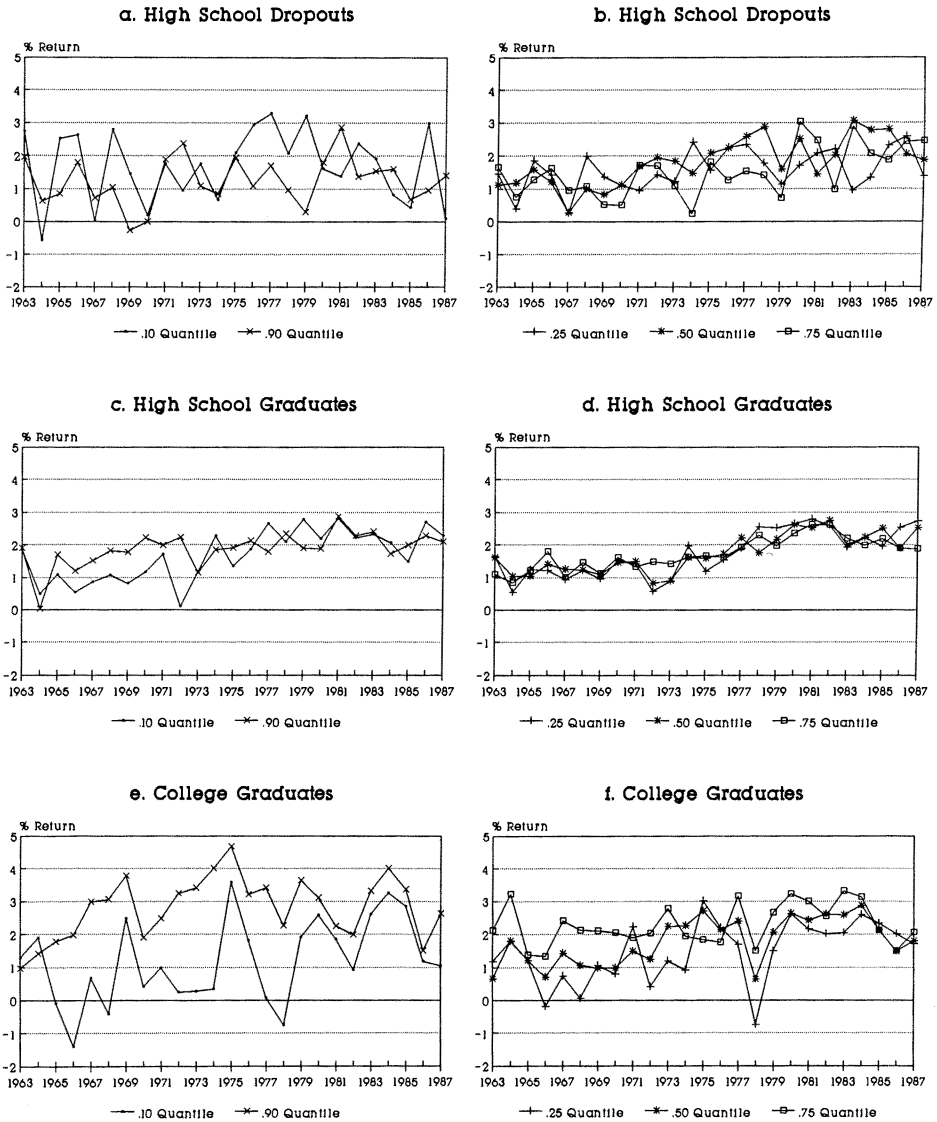


FIGURE 10.—16-group model—return to experience for experienced workers (15 years of experience), by quantiles.

Similar to the return to education, the patterns of change in the return to experience are clearer for the high school and college graduates than for the high school dropouts. Year-to-year fluctuations are exceedingly large (in part due to sampling variation) for the last group, with no clear ordering across the five quantiles.

A feature common with the one-group model is that changes are smaller and more gradual for the experienced workers than for the new entrants. For the latter a large increase during the 1960's was followed by a large decrease during the 1970's, at all quantiles. In contrast, for the experienced workers a small general upward trend is apparent at all education levels throughout the 1960's and 1970's.

In summary the main results are: (a) the returns to experience for the new entrants and experienced workers groups exhibit different patterns of behavior, both over time and across quantiles; (b) the returns at the five quantiles converged during the 1980's for both experience groups; and (c) the returns for the high school dropouts behave erratically at all experience levels and across all quantiles.

4.3. *Changes in Within-Group Wage Inequality*

The changes in returns to education and experience had a major impact on wage inequality, especially during the second half of the sample period. The pattern has been previously documented by several authors (e.g., Blackburn, Bloom, and Freeman (1990), Juhn, Murphy, and Pierce (1993), Blackburn and Bloom (1987), and Katz and Revenga (1989)). The results presented here are somewhat more detailed.

Since the quantile regression technique allows one to trace the entire conditional wage distribution, it provides an extremely useful tool for examining changes in the shape of that conditional distribution. I consider two measures of within-group inequality, defined as the difference between the relevant conditional quantiles: (a) the *.90-.10 spread*; and (b) the *.75-.25 spread*. To better examine changes in the shape of the distribution, each of these measures is decomposed into two half-spreads: (a) into a *.90-.50* and a *.50-.10* half spread, and (b) similarly into a *.75-.50* and a *.50-.25* half-spread.

There is an essential difference between the two models analyzed in this study. (Results are presented, however, only for the 16-group model.) For the one-group model the measures of inequality for distinct skill groups reflect disparity only in the covariate values, while for the 16-group model they reflect differences in both the covariate values and the estimated parameters. The measures from the one-group model are expected therefore to show less variability than from the 16-group model. One should be cautious in interpreting the results since the year-to-year fluctuations are quite large.

Figures 11 and 12 graph the within-group wage inequality measures for the new entrants group. Figure 11 presents the *.90-.10* spread and its two half-spreads, along with an alternative measure of dispersion, 2.56σ . The latter

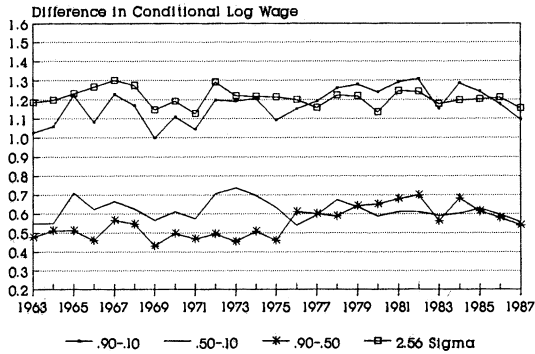
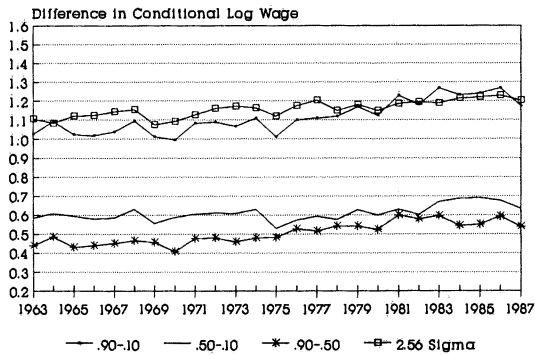
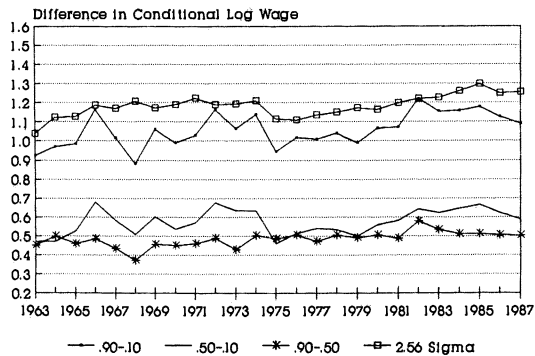
a. High School Dropouts**b. High School Graduates****c. College Graduates**

FIGURE 11.—16-group model—within-group wage inequality, .90-.10 spread and σ , for new entrants (5 years of experience).

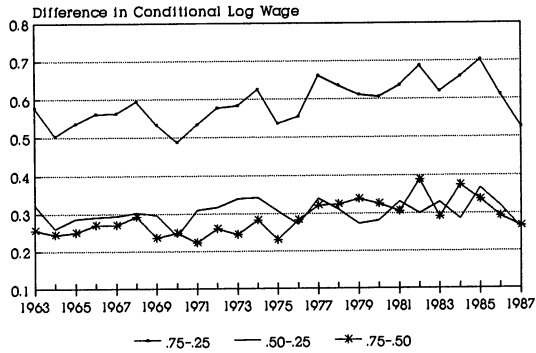
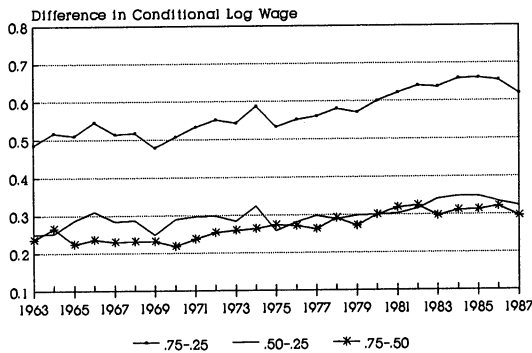
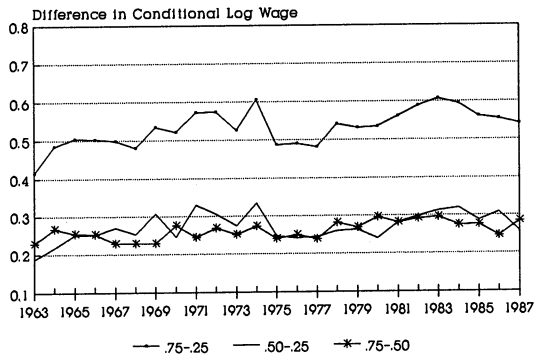
a. High School Dropouts**b. High School Graduates****c. College Graduates**

FIGURE 12.—16-group model—within-group wage inequality, .75–.25 spread, for new entrants (5 years of experience).

measure is obtained from the residual series of a least-squares regression in the education-experience group, using the same set of variables as in the quantile regressions for the group.³⁵ The .75-.25 spread is depicted similarly in Figure 12. Figures 13 and 14 graph the same measures for the experienced workers group, while Figures 15 and 16 are for the prime earners.

For most skill groups the .90-.10 spread and the 2.56σ measure changed in similar patterns, although the former measure has larger year-to-year fluctuations. Some differences in the relationship between the two inequality measures are observed across skill groups.

For the new entrants group the 2.56σ measure is larger than the .90-.10 spread for the college graduates throughout the sample period. For the high school graduates and especially for the high school dropouts, the 2.56σ spread is smaller than the .90-.10 spread only during the earlier part of the period. This indicates that for the less educated workers the log wage distribution was not as widely spread as is implied by a normal distribution during the earlier part of the sample period, but it was more widely spread toward the end of the period.

There is no clear separation between the two measures for the experienced workers and prime earners groups. The log wage distribution is not always more widely spread than the implied normal distribution. Nevertheless, the two measures do differ in almost every year, especially for the prime earners group. For this last group the 2.56σ measure appears to be smaller than the .90-.10 spread for the least and most educated groups toward the end of the period.

I summarize below the results for each of the three experience groups and the three education groups within each of these experience groups.

New Entrants Experience Group

Changes in wage inequality for this experience group were quite different across the three education groups. The overall change in the .90-.10 spread between 1963 and 1987 was only 6.6% for the high school dropouts, but much larger for the high school and the college graduates, namely 14.2% and 18%. Changes for the high school dropouts were also more erratic than for the other two groups, which experienced a continuous increase of 15.7% and 15.3% from 1975 to 1987. Also, for these last two groups the .50-.10 spreads are considerably higher than the .90-.50 spreads, with the difference increasing toward the end of the period. For the high school dropouts these two half-spreads are indistinguishable from 1976 on.

The .75-.25 measures of inequality for the new entrants exhibit more pronounced differences among the three education groups. The high school graduates experienced the most steady increase in the .75-.25 spread of 27.9%

³⁵ Specifically, the standard deviation σ is computed from least-squares regressions in each of the 16 skill groups. Each such regression includes the same variables as the corresponding quantile regressions. For the j th groups ($j = 1, \dots, 16$) $\hat{\sigma}_j^2 = (1/N_j) \sum_{i=1}^{N_j} e_{ji}^2$, where $e_{ji} = y_{ji} - x'_{ji} \hat{\beta}_{LS}$, N_j is the number of observations, and $\hat{\beta}_{LS}$ is the least-squares estimate. If the error term is normally distributed, then 2.56σ is the same as the .90-.10 spread. I wish to thank an anonymous referee for suggesting this measure.

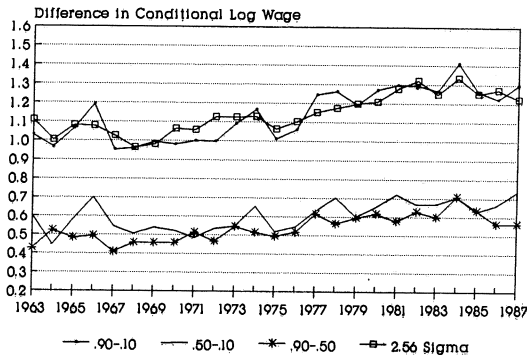
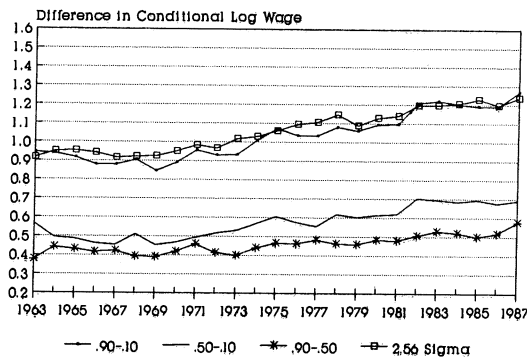
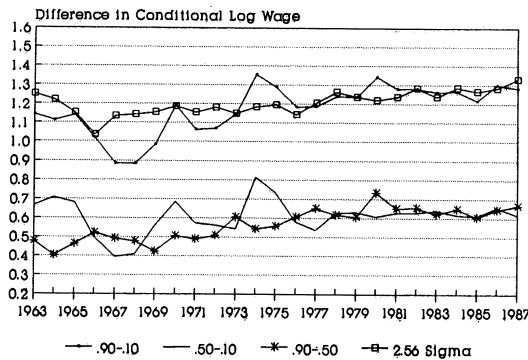
a. High School Dropouts**b. High School Graduates****c. College Graduates**

FIGURE 13.—16-group model—within-group wage inequality, .90-.10 spread and σ , for experienced workers (15 years of experience).

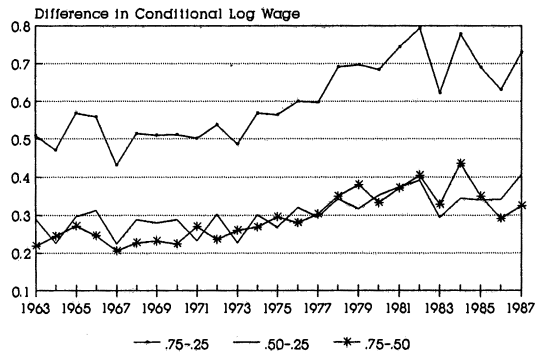
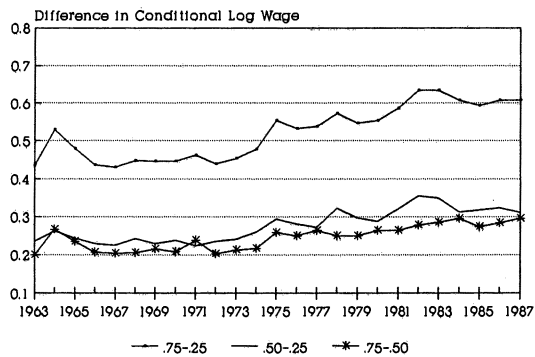
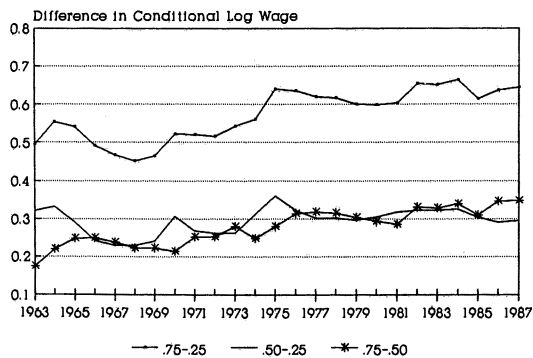
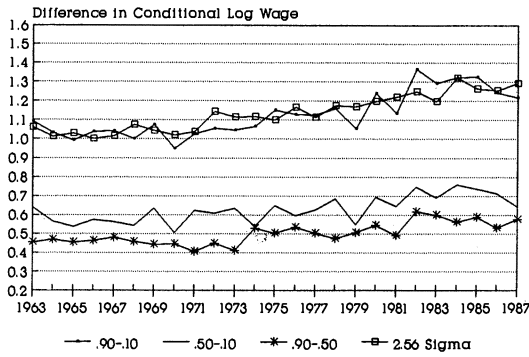
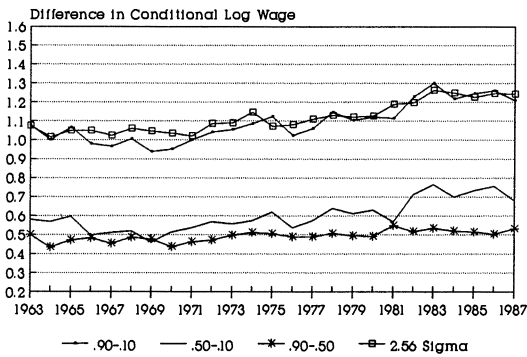
a. High School Dropouts**b. High School Graduates****c. College Graduates**

FIGURE 14.—16-group model—within-group wage inequality, .75-.25 spread, for experienced workers (15 years of experience).

a. High School Dropouts



b. High School Graduates



c. College Graduates

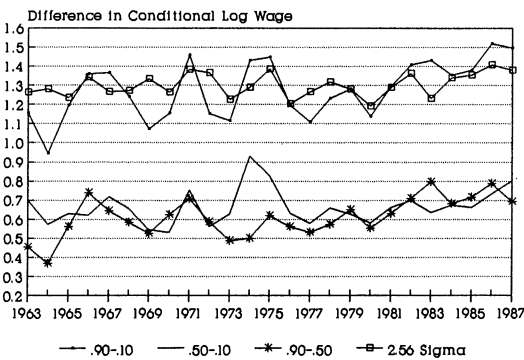


FIGURE 15.—16-group model—within-group wage inequality, .90-.10 spread and σ , for prime earners (25 years of experience).

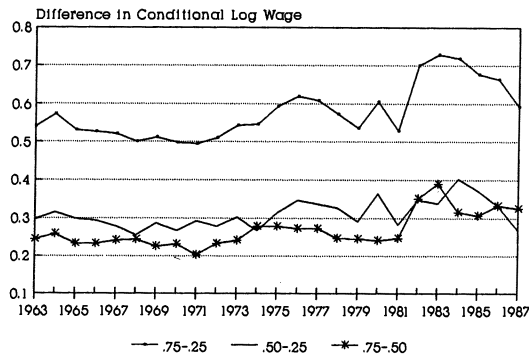
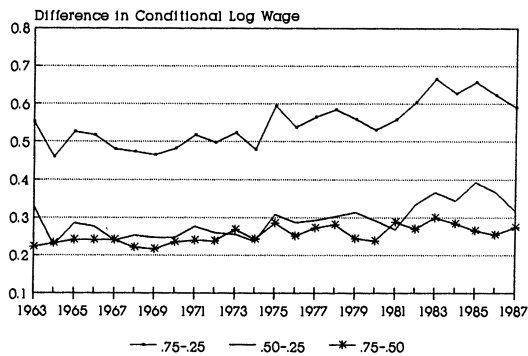
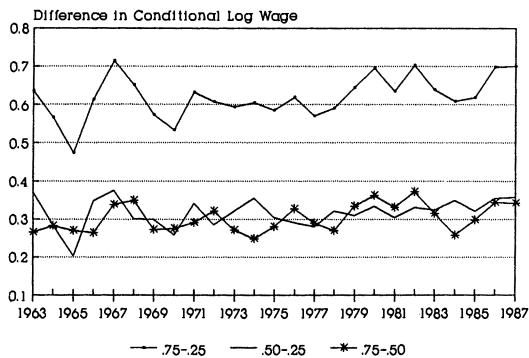
a. High School Dropouts**b. High School Graduates****c. College Graduates**

FIGURE 16.—16-group model—within-group wage inequality, .75-.25 spread, for prime earners (25 years of experience).

between 1969 and 1987. This spread rose also by 24.6% for the college graduates, but it declined considerably from 1983 on. For the high school dropouts the .75-.25 spread rose by 12.6% over the same period but it fluctuated enormously, with short periods (3-4 years) of changes in the continuous trend.

Note that the .75-.50 and .50-.25 half-spreads are about the same for all three education groups. This suggests that changes in the overall .75-.25 spread were induced by symmetric changes at both sides of the distribution. In contrast, for the .90-.10 spread the lower half spreads tend to be larger than the upper half spreads, toward the end of the sample period, at least for the high school graduates and college graduates.

Experienced Workers Group

For the experienced workers the overall increase in the .90-.10 spread, between 1963 and 1987, were 26.0%, 34.2%, and 12.2%, for the high school dropouts, high school graduates, and college graduates, respectively. However, while the .90-.10 spread increased continuously for the high school graduates through most of the period, the increase was small from 1981 for the high school dropouts, and it even decreased slightly from 1975 for the college graduates.

The two half-spreads also exhibit different patterns of change. For the high school graduates the .50-.10 spread is consistently above the .90-.50 spread and increases faster, while for the other two groups the two half-spreads are not far apart, with the .50-.10 spread fluctuating more than the .90-.50 spread.

Similar patterns of change are observed in the .75-.25 spread. Overall it increased between 1963 and 1987 by 43.5%, 39.5%, and 30.2% for the high school dropouts, the high school graduates, and the college graduates, respectively. Similar to the .90-.10 half-spreads, the .50-.25 spread is consistently above the .75-.50 spread only for the high school graduates while for the other two groups the two half-spreads fluctuate about each other.

Prime Earners Group

As with the other two experience groups there are significant differences among the three education groups. Throughout the entire period the .90-.10 spread increased by 26.6% for the college graduates group, with large swings over short periods of time. For the other two education groups the .90-.10 spread and its two half-spreads changed in a similar pattern. The .50-.10 spread was above the .90-.50 and increased faster toward the end of the period for both groups. The overall .90-.10 spread increased between 1963 and 1987 by 11.6% and 12.2% for the high school dropouts and high school graduates groups, respectively.

Significant changes occurred in the .75-.25 spread for the high school dropouts and the high school graduates. From 1979 to 1983 the .75-.25 spread increased significantly by 34.6% and 28.8% for these two groups, respectively,

while from 1983 to 1987 it declined by 16.7% and 11.9%. For both groups the .50-.25 half-spread fluctuated more than the .75-.50 half spread. Overall the .75-.25 spread increased from 1963 to 1987 by 16.0%, 27.1%, and 22.3% for the three education groups, respectively.

To summarize, the changes in the .90-.10 spreads, and even more so in the .75-.25 spreads, indicate that substantial changes occurred in the shape of the wage distribution. In particular, for most of the early period the increased wage inequality arose primarily from longer tails at both ends of the wage distribution. Toward the end of the period the increased inequality for most skill groups reflects a longer tail at the low end of the wage distribution. The pattern of changes in the .90-.10 and the .75-.25 spreads, and especially in their half-spreads, differs significantly across the various skill groups. Within-group wage inequality has not changed much over the entire period for the most skilled and least skilled workers, but it has increased significantly for those workers in between.

The results for the one-group model are not in general consistent with the 16-group model; in particular, the former model does not seem to capture the changes for the high school dropouts. This implies that a single-equation model needs to allow for interactions among the covariates and also requires a more flexible functional form in education and experience. The present one-group model is unable to capture changes in the wage structure in general, and in the returns to skills in particular, across groups with considerable differences in education and experience attainments.

Such a model is investigated in Buchinsky (1994a). In this study a flexible functional form, which allows also for nontrivial interaction among all covariates, is obtained from the well known Box-Cox transformation model and is fitted to the full data. This Box-Cox model captures some of the important features observed in the 16-group model, but the results are not completely consistent with the 16-group model.

5. SUMMARY OF THE EMPIRICAL RESULTS AND CONCLUSIONS

"On the average" has never been a satisfactory statement with which to conclude a study on heterogeneous populations. Characterization of the conditional mean constitutes only a limited aspect of possibly more extensive changes involving the entire distribution.

This paper attempts to remedy this deficiency by employing quantile regression techniques that characterize the entire conditional distribution. Specifically, changes in the return to education and experience are examined at five different quantiles: .10, .25, .50, .75, and .90. In addition, two measures of inequality are defined, the .90-.10 and .75-.25 spreads, which are effective descriptive tools in analyzing changes in within-group wage inequality.

Two models are employed: (a) a one-group linear model; and (b) a 16-group linear model. In the former a single linear equation of log weekly wage is

TABLE VI
SUMMARY STATISTICS FOR THE ONE-GROUP MODEL AND 16-GROUP MODEL

Year	Total Number of Observations		Minimum Distance Estimation (One-Group Model)					Test for Equality
			% Included by Quantiles	No. of Cells Used by Quantiles				
	16 Group	One-Group	.75	.90	< .75	.75	.90	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1963	11633	8856	100.0	100.0	205	205	205	35.9
1964	11807	9000	100.0	100.0	193	193	193	46.9
1965	25455	20926	100.0	100.0	423	423	423	80.2
1966	15825	11608	100.0	100.0	298	298	298	46.9
1967	24664	20380	100.0	100.0	428	428	428	73.0
1968	25006	20729	100.0	100.0	433	433	433	60.4
1969	23909	19757	100.0	99.9	422	422	421	65.9
1970	23499	19234	100.0	100.0	400	400	400	71.9
1971	22955	19035	100.0	100.0	408	408	408	81.1
1972	22687	18804	100.0	100.0	396	396	396	56.0
1973	22231	18335	100.0	99.8	383	383	382	47.1
1974	20159	16703	100.0	99.9	365	365	364	73.4
1975	24306	20520	100.0	99.4	414	414	410	48.7
1976	29025	25275	100.0	98.8	451	451	441	35.3
1977	28944	25248	100.0	98.4	450	450	440	34.9
1978	28911	25028	100.0	97.9	435	435	421	36.3
1979	33603	29918	99.6	95.7	478	474	454	64.2
1980	33208	29389	99.7	93.7	465	462	430	73.2
1981	28891	25015	100.0	99.4	428	428	422	46.2
1982	27759	24030	99.9	97.5	407	406	394	62.9
1983	28401	24588	99.8	97.3	406	405	392	45.9
1984	29184	25294	100.0	99.4	404	404	400	29.3
1985	28774	24810	100.0	99.6	392	392	390	45.1
1986	28547	24597	100.0	98.6	389	389	382	76.5
1987	29005	25179	100.0	98.7	400	400	394	77.7

Note: Column (2) gives the number of observations used for the 16-group model. Column (3) gives the number of observations used for the MD estimation of the one-group model, using only cells with 15 or more observations and 7 or more years of education. Columns (4) and (5) report the percentage of observations actually used (out of the number in column (3)) due to censoring, while columns (6), (7), and (8) report the number of cells used. For all the regressions at quantiles below .75 no censoring problem arose. The number of cells used for these regressions are reported in column (6). The test statistic in column (9) is $T = n(\hat{\beta} - G_R \hat{\beta}_R)A^{-1}(\hat{\beta} - G_R \hat{\beta}_R)$, where the notation is the same as in footnote 27. Under the null hypothesis $T \sim \chi^2(16)$.

estimated for the entire sample, while in the latter one such equation is estimated separately for 16 different education-experience groups.

The returns to education and experience from the one-group model exhibit similar patterns of growth across the different quantiles, but are substantially different in their levels. In general, the returns to education are higher at the higher quantiles, while the returns to experience are higher at the lower quantiles, especially during the early 1970's. The returns to experience at all quantiles converge toward the end of the sample period.

The results obtained for the 16-group model differ somewhat and are more pointed: (a) the overall increase in the return to education is primarily due to the increase in the return to college education for the new entrants and experienced workers, and to the increase in the return to high school education

for the new entrants; (b) the returns to education are higher, in general, at the higher quantiles for the new entrants, and lower at the higher quantiles for experienced workers and prime earners; (c) changes in the returns to experience are not uniform across quantiles or across the different skill levels; being higher at the lower quantiles for the new entrants, but higher at the higher quantiles for the experienced workers; and (d) the difference in the return to experience for the new entrants is large during the first half of the sample period but smaller during the 1980's.

Investigation of the two measures of inequality, the .90-.10 spread and the .75-.25 spread, reveals that: (a) the increase in wage inequality seems to be more systematic for the highly skilled groups, and more erratic for the less-skilled workers; (b) large changes in the .75-.25 spreads suggest fundamental changes in the shape of the wage distribution, not merely in the tails; (c) changes in inequality come from longer tails at both ends of the wage distribution; although toward the end of the sample period the increased inequality for most skill groups seems to have come from longer lower tails.

Although the analyses here quantify the magnitude of changes in wage inequality and in the returns to skills at different quantiles, underlying causes are not explicitly identified. The importance of the results presented here, especially pertaining to the returns to education and experience, lies not in any immediate implications for the existing hypotheses proposed in the literature, but rather in their providing a basis for hypothesis testing within the context of a particular model. In any such model implications for the quantile estimates can be derived and compared with those obtained in this study.

The results presented here are consistent with those reported in the literature for studies which employ mean type regressions. Although the returns at the five quantiles are different, the patterns of change are similar across all quantiles for almost all skill groups. One important finding of this study is that whatever the causes that induced changes in the returns to education and experience, the effects were similar at all points of the wage distribution.

Part of the existing literature has attributed the changes in the wage structure to technological changes, institutional changes, and the decline in unionization. This might lead one to believe that the patterns of change in the returns to skills should differ at different quantiles of the wage distribution. Such hypotheses are not necessarily wrong, but for meaningful comparison with the results reported here one needs to derive the implications implied by a specific model. This is beyond the scope of the current study.³⁶ The results in this paper strongly suggest, however, that any formal structural model should allow for potentially diverse changes at different points in the wage distribution.

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³⁶ Some of the implications of specific models are discussed in Buchinsky (1994a).

APPENDIX

IMPUTATION METHOD FOR "WEEKS WORKED LAST YEAR" PRIOR TO 1975

The basic assumption underlying the imputation method is that the distribution of weeks worked per year is stationary over the sample period for every race-age-education (characteristic) group. Altogether there are 2014 such groups.

A 2014×53 matrix is created in which each row corresponds to a group, and each column corresponds to the number of weeks worked in the previous year. The form of this matrix is:

$$\begin{array}{c} \text{Group 1:} \\ \vdots \\ \text{Group 2014:} \end{array} \begin{array}{c} \overbrace{\begin{array}{ccccc} & \text{Weeks worked last year} & \\ 0 & 1 & 2 & \cdots & 52 \end{array}} \\ \left(\begin{array}{ccccc} f_{1,0} & f_{1,1} & f_{1,2} & \cdots & f_{1,52} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ f_{2014,0} & f_{2014,1} & f_{2014,2} & \cdots & f_{2014,52} \end{array} \right) \equiv F. \end{array}$$

Three such matrices, F_{75} , F_{76} , and F_{77} , were computed, for the three years 1975, 1976, and 1977, respectively. To reduce the variation between years which are not due to sample design, the three matrices are added together to obtain the *basic frequency matrix* F ($F = F_{75} + F_{76} + F_{77}$). The elements of F , denoted by f_{ij} ($i = 1, \dots, 2014$; $j = 0, \dots, 52$), are the number of people in the i th group who worked j hours in the previous year.

The bracketed version of the "weeks worked last year" variable comes in seven categories: (1) 0 weeks worked last year; (2) 1–13 weeks; (3) 14–26 weeks; (4) 27–39 weeks; (5) 40–47 weeks; (6) 48–49 weeks; and (7) 50–52 weeks. Also created is a bracketed version of F , say F^B , which is a 2014×7 matrix, with elements f_{ij}^B in F^B formed as follows:

$$f_{i1}^B = f_{i,0}, \quad f_{i2}^B = \sum_{l=1}^{13} f_{i,l}, \dots, \quad f_{i7}^B = \sum_{l=50}^{52} f_{i,l}.$$

Note that the total number of people in the i th row of F is $f_i = \sum_{j=1}^7 f_{ij}^B = \sum_{k=0}^{52} f_{ik}$.

The imputation is done in two steps for each bracketed category of every group: (a) the number of people who worked a particular number of weeks, in each category of the bracketed version variable, is imputed from the corresponding row in F ; and (b) the values imputed in (a) are assigned to the individuals in the bracketed category.

Denote by n_i the total number of people in the i th group ($i = 1, \dots, 2014$), and let n_{ij} denote the number of individuals in the j th bracket of the i th group. Note that $n_i = \sum_{j=1}^7 n_{ij}$. Also, let $n_{ij}(m)$ be the number of people in the j th category of the i th group who worked m weeks last year ($m = k, \dots, k+l$). The quantity $n_{ij}(m)$ is imputed then for two possible cases:

(i) $n_{ij} > 0$, but $f_{ij}^B = 0$: The median point of the j th category is assigned to all individuals in the i th group. (For example, if $n_{i7} = 5$ and $f_{i7}^B = 0$, then the value 51 is assigned to all 5 individuals in the 7th category of the i th group.)

(ii) $n_{ij} > 0$ and $f_{ij}^B > 0$: The following simple formula is applied:

$$n_{ij}(m) = R\left(\frac{n_{ij} f_{im}}{f_{ij}^B}\right), \quad m = k, \dots, k+l,$$

where $R(\lambda)$ denotes the closest rounded number of λ .

After imputing the number of people who worked m weeks, for each m , a value of "weeks worked last year" is assigned to each individual. This value must be consistent with the bracketed value. In each bracketed category the largest imputed number is assigned to the individual with the highest income from wages and salaries, the second largest value to the individual with the second highest income, etc. For example, suppose $n_{i7} = 3$ and we imputed $n_{i7}(50) = 0$, $n_{i7}(51) = 2$, and $n_{i7}(52) = 1$; then the individual with the highest income is assigned the value 52, while the remaining two individuals are each assigned the value 51.

REFERENCES

- BARRODALE, I., AND F. ROBERTS (1973): "An Improved Algorithm for Discrete l_1 Linear Approximation," *SIAM Journal of Numerical Analysis*, 10, 839–848.
- BARTELS, R., AND A. CONN (1980): "Linearly Constrained Discrete l_1 Problems," *Association for Computing Machinery Transactions on Mathematical Software*, 6, 594–608.
- BICKEL, P., AND D. FREEDMAN (1981): "Some Asymptotic Theory for the Bootstrap," *Annals of Statistics*, 9, 1196–1217.
- BLACKBURN, M., AND D. BLOOM (1987): "Earnings and Income Inequality in the United States," *Population and Development Review*, 13, 575–609.
- BLACKBURN, M., D. BLOOM, AND R. FREEMAN (1990): "The Declining Position of Less Skilled American Males," in *A Future of Lousy Jobs?*, ed. by G. Burtless. Washington, D.C.: Brookings Institution, 31–67.
- (1991): "Changes in Earning Differentials in the 1980's: Concordance, Convergence, Causes, and Consequences," Discussion Paper Series No. 554, Department of Economics, Columbia University.
- BOUND, J., AND G. JOHNSON (1991): "Wages in the United States During the 1980's and Beyond," in *Workers and Their Wages*, ed. by M. Kosters. Washington, D.C.: The AEI Press, 77–103.
- (1992): "Changes in the Structure of Wages During the 1980's: An Evaluation of Alternative Explanations," *American Economic Review*, 82, 371–392.
- BUCHINSKY, M. (1991): "Methodological Issues in Quantile Regression," Chapter 1 of *The Theory and Practice of Quantile Regression*, Ph.D. dissertation, Harvard University.
- (1992): "Bootstrapping Quantile Regression Model," unpublished manuscript, Yale University.
- (1994a): "Quantile Regression, Box-Cox Transformation Model and Changes in the Returns to Schooling and Experience," *Journal of Econometrics*, forthcoming.
- (1994b): "A Monte Carlo Study of the Asymptotic Covariance Matrix Estimators for Quantile Regression Estimators," unpublished manuscript, Yale University.
- CARD, D., AND A. KRUEGER (1992a): "Does School Quality Matter? Returns to Education and the Characteristics of Public Schools in the United States," *The Journal of Political Economy*, 100, 1–40.
- (1992b): "School Quality and Black-White Relative Earnings: A Direct Assessment," *The Quarterly Journal of Economics*, 107, 151–200.
- CHAMBERLAIN, G. (1991): "Quantile Regression, Censoring, and the Structure of Wage," Harvard Institute of Economic Research, Discussion Paper No. 1558.
- DAVIS, S., AND J. HALTIWANGER (1991): "Wage Dispersion Between and Within U.S. Manufacturing Plants, 1962–1986," *Brookings Papers on Economic Activity: Microeconomics*, 115–180.
- EFRON, B. (1979): "Bootstrap Methods: Another Look at the Jackknife," *Annals of Statistics*, 7, 1–26.
- (1982): *The Jackknife, the Bootstrap and Other Resampling Plans*. Philadelphia: Society for Industrial and Applied Mathematics.
- FREEDMAN, D. (1981): "Bootstrapping Regression Models," *The Annals of Statistics*, 9, 1218–1228.
- FREEDMAN, D., AND S. PETERS (1984a): "Bootstrapping an Econometric Model: Some Empirical Results," *Journal of the American Statistical Association*, 79, 97–106.
- (1984b): "Bootstrapping a Regression Model: Some Empirical Results," *Journal of Business Economics and Statistics*, 2, 150–158.
- FREEMAN, R. (1991): "How Much Has De-Unionization Contributed to the Rise in Male Earnings Inequality?" NBER Working Paper, No. 3826.
- HAHN, J. (1991): "Bootstrapping a General Class of M -Estimators," unpublished manuscript, Harvard University.
- (1992): "Bootstrapping Quantile Regression Models," unpublished manuscript, Harvard University.
- HECKMAN, J., AND R. ROBB (1985): "Using Longitudinal Data to Estimate Age, Period and Cohort Effects in Earnings Equations," in *Cohort Analysis in Social Research*, ed. by W. M. Mason and S. E. Feinberg. New York: Springer-Verlag.
- HUBER, P. (1967): "The Behavior of Maximum Likelihood Estimates Under Nonstandard Conditions," *Proceedings of the Fifth Berkeley Symposium*, 4, 221–223.
- (1981): *Robust Statistics*. New York: Wiley.
- JOHN, C., K. MURPHY, AND B. PIERCE (1993): "Wage Inequality and the Rise in Returns to Skill," *Journal of Political Economy*, 101, 410–442.

- KATZ, L., AND K. MURPHY (1992): "Changes in the Relative Wages, 1963–87: Supply and Demand Factors," *The Quarterly Journal of Economics*, 107, 35–78.
- KATZ, L., AND A. REVENGA (1989): "Changes in the Structure of Wages: The U.S. Versus Japan," *Journal of Japanese and International Economics*, 3, 522–553.
- KOENKER, R., AND G. BASSETT (1978): "Regression Quantiles," *Econometrica*, 46, 33–50.
- KRUEGER, A. (1991): "How Computers Have Changed the Wage Structure: Evidence from Microdata, 1984–1989," mimeo, Princeton University.
- LEVY, F., AND R. MURNANE (1992): "U.S. Earnings Levels and Earnings Inequality: A Review of Recent Trends and Proposed Explanation," *Journal of Economic Literature*, 30, 1333–1381.
- LO, S.-H. (1989): "On Some Representation of the Bootstrap," *Probability Theory and Related Fields*, 82, 411–418.
- MANSKI, C. (1988): *Analog Estimation Methods in Econometrics*. New York: Chapman and Hall.
- MINCER, J. (1974): *Schooling, Experience and Earnings*. New York: National Bureau of Economic Research.
- (1991): "Human Capital, Technology, and the Wage Structure: What do Time Series Show?" NBER Working Paper No. 3581.
- MITCHELL, D. (1989): "Wage Pressures and Labor Shortages: The 1960s and 1980s," *Brookings Papers on Economic Activity: Microeconomics*, 2, 191–232.
- MURPHY, K., AND F. WELCH (1989): "Recent Trends in Real Wages: Evidence from Household Data," Program in Applied Econometrics, Unicon Research Corporation, Los Angeles, CA, Discussion Paper No. 24.
- (1990): "Empirical Age-Earning Profiles," *Journal of Labor Economics*, 8, 202–229.
- (1991): "The Role of International Trade in Wage Differentials," in *Workers and Their Wages*, ed. by M. Koster. Washington, D.C.: The AEI Press, 39–69.
- (1992): "The Structure of Wages," *The Quarterly Journal of Economics*, 107, 285–326.
- NEWWEY, W., AND J. POWELL (1990): "Efficient Estimation of Linear and Type I Censored Regression Models Under Conditional Quantile Restrictions," *Econometric Theory*, 6, 295–317.
- PORTNOY, S., AND R. KOENKER (1989): "Adaptive L -Estimation for Linear Models," *The Annals of Statistics*, 17, 362–381.
- POWELL, J. (1984): "Least Absolute Deviation Estimation for the Censored Regression Model," *Journal of Econometrics*, 25, 303–325.
- (1986): "Censored Regression Quantiles," *Journal of Econometrics*, 32, 143–155.
- SINGH, K. (1981): "On the Asymptotic Accuracy of Efron's Bootstrap," *The Annals of Statistics*, 9, 1187–1195.
- WOHLSTETTER, A., AND S. COLEMAN (1972): "Race Differences in Income," in *Racial Discrimination in Economic Life*, ed. by A. Pascal. Lexington, Mass: D. C. Heath, 3–81.
- WOMERSLEY, R. S. (1986): "Censored Discrete Linear l_1 Approximation," *SIAM, Journal of Scientific and Statistical Computing*, 7, 105–122.